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Pyrido[2,3-e]oxazine derivatives as agricultural chemicals

Abstract

The present invention relates to picolinic acid derivatives that are useful in treating fungal diseases, particularly fungal diseases of agricultural crop plants.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a U.S. National Stage entry under 35 U.S.C. § 371 of International Application No. PCT/EP2021/051814, filed on Jan. 27, 2021, designating the United States of America and published in English on Aug. 5, 2021, which in turn claims priority to G.B. Application No. 2001181.3, filed on Jan. 28, 2020, each of which is hereby incorporated by reference in its entirety.

(2) The present invention relates to picolinic acid derivatives that are useful in treating fungal disease.

(3) Given the global increase in demand for food, there is an international need for new treatments to reduce food crop losses to disease, insects and weeds. Over 40% of crops are lost before harvest, and 10% post harvest, worldwide. Losses have actually increased since the mid-1990s.

(4) A new threat contributing to this is the emergence of chemical-resistant organisms, for example, glyphosate-resistant weeds in the USA and strobilurin-resistant strains of *septoria* fungal species.

(5) Recent research also suggests that the geographical spread of many crop pests and diseases is increasing, possibly as a result of global warming.

(6) An aim of certain embodiments of the present invention is to provide pesticides (e.g. fungicides) which have activity either non-selectively, i.e. broad spectrum activity, or which are active specifically against selective target organisms.

(7) An aim of certain embodiments of the present invention is to provide compounds which are less persistent in the environment after use than prior art compounds. Alternatively or additionally, the compounds of the present invention may be less prone to bioaccumulation once in the food chain than prior art compounds.

(8) Another aim of certain embodiments of the invention is to provide compounds which are less harmful to humans than prior art compounds. Alternatively or additionally, the compounds of the invention may be less harmful than prior art compounds to one or more of the following groups: amphibians, fish, mammals (including domesticated animals such as dogs, cats, cows, sheep, pigs, goats, etc.), reptiles, birds, and beneficial invertebrates (e.g. bees and other insects, or worms), beneficial nematodes, beneficial fungi and nitrogen-fixing bacteria.

(9) The compounds of the invention may be as active as or more active than prior art compounds. They may have activity against organisms which have developed a resistance to prior art compounds. However, the present invention may also concern compounds which have a lower or similar level of activity relative to that of the prior art compounds. These lower activity compounds are still effective as fungicides but may have other advantages relative to existing compounds such as, for example, a reduced environmental impact.

(10) The compounds of the invention may be more selective than prior art compounds, i.e. they may have better, similar or even slightly lower activity than prior art compounds against target species but have a significantly lower activity against non-target species (e.g. the crops which are being protected).

(11) Certain embodiments of the invention provide compounds that achieve one or more of the above aims. The compounds may be active in their own right or may metabolise or react in aqueous media to yield an active compound.

SUMMARY OF THE INVENTION

(12) In a first aspect of the invention is provided a compound of formula I, or an agronomically acceptable salt or N-oxide thereof:

(13) ##STR00001##

(14) X.sup.1 and X.sup.2 are each independently selected from carbon and nitrogen; Y.sup.1 is independently selected from O and S; =Y.sup.2 is independently selected from =O and =S; Z.sup.1 is independently absent or is selected from C(O)O, OC(O), O, S, S(O), S(O).sub.2, C(O)NR.sup.5, NR.sup.5C(O), S(O).sub.2NR.sup.5, NR.sup.5S(O).sub.2, S(O)NR.sup.5, NR.sup.5S(O), CR.sup.6R.sup.7, C(O), C(S), C=NOR.sup.8, C.sub.1-C.sub.3-alkylene and NR.sup.5; R.sup.1 and R.sup.12 are each independently at each occurrence selected from C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.3-C.sub.6-cycloalkyl, halogen, nitro, OR.sup.9, SR.sup.10, OS(O).sub.2R.sup.10, S(O).sub.2R.sup.10, C(O)OR.sup.10, C(O)NR.sup.10R.sup.10, C(O)R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, and NR.sup.10R.sup.11; R.sup.2 is independently at each occurrence selected from C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.3-C.sub.6-cycloalkyl, halogen, nitro, OR.sup.9, SR.sup.10, OS(O).sub.2R.sup.10, S(O).sub.2R.sup.10, C(O)OR.sup.10, C(O)NR.sup.10R.sup.10, C(O)R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, NR.sup.10R.sup.11 and phenyl optionally substituted with from 1 to 5 R.sup.12 groups; R.sup.3 is independently at each occurrence selected from C.sub.3-C.sub.8-alkyl and C.sub.0-C.sub.3-alkylene-R.sup.3a; wherein R.sup.3a is independently at each occurrence selected from phenyl, 5- or 6-membered heteroaryl, 5-, 6-, 7- or 8-membered heterocycloalkyl and C.sub.3-C.sub.8-cycloalkyl; said heterocycloalkyl or cycloalkyl group being monocyclic or bicyclic; said heteroaryl or phenyl group being optionally substituted with from 1 to 5 R.sup.12 groups or said heterocycloalkyl or cycloalkyl group being optionally substituted with from 1 to 4 R.sup.13 groups; said heterocycloalkyl or cycloalkyl group being optionally fused to phenyl or a 5- or 6-membered heteroaryl, said heteroaryl or phenyl group being optionally substituted with from 1 to 4 R.sup.12 groups; R.sup.4 is independently at each occurrence selected from: C.sub.1-C.sub.6-alkyl, C.sub.3-C.sub.6-cycloalkyl and C.sub.1-C.sub.6-haloalkyl; R.sup.5 and R.sup.10 are each independently at each occurrence selected from: H, C.sub.3-C.sub.6-cycloalkyl, C.sub.1-C.sub.6-alkyl and benzyl; or where two R.sup.10 groups are attached to the same nitrogen atom, said R.sup.10 groups, together with said nitrogen atom form a 4-, 5-, 6- or 7-membered heterocycloalkyl ring; R.sup.6 is independently at each occurrence selected from: H, C.sub.3-C.sub.6-cycloalkyl, C.sub.1-C.sub.6-alkyl, phenyl and 5- or 6-membered heteroaryl; R.sup.7 is independently at each occurrence selected from: H, halo and OR.sup.8; R.sup.8 is each independently at each occurrence selected from: H, C.sub.3-C.sub.6-cycloalkyl, C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.3-alkylene-R.sup.8a; wherein R.sup.8a is independently at each occurrence selected from phenyl and 5- or 6-membered heteroaryl; R.sup.9 is independently at each occurrence selected from: H, C.sub.1-C.sub.6-alkyl, C.sub.0-C.sub.3-alkylene-C.sub.3-C.sub.6-cycloalkyl, C(O)—C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.6-haloalkyl; R.sup.11 is independently at each occurrence selected from: H, C.sub.1-C.sub.6-alkyl, C(O)—C.sub.1-C.sub.6-alkyl and S(O).sub.2—C.sub.1-C.sub.6-alkyl; or where an R.sup.10 group and an R.sup.11 group are attached to the same nitrogen atom, said R.sup.10 and R.sup.11 groups, together with said nitrogen atom form a 4-, 5-, 6- or 7-membered heterocycloalkyl ring; R.sup.13 is independently at each occurrence selected from: =O, =S, C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.3-C.sub.6-cycloalkyl; halogen, nitro, OR.sup.9, SR.sup.10, OS(O).sub.2R.sup.10, S(O).sub.2R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, and NR.sup.10R.sup.11; y is independently an integer selected from 0, 1 and 2; m is independently an integer selected from 0, 1, 2 and 3; and wherein any aforementioned alkyl, alkylene, alkenyl, cycloalkyl, heterocycloalkyl (including where two R.sup.10 groups or an R.sup.10 group and an R.sup.11 group together with a nitrogen to which they are attached form a heterocycloalkyl ring), alkynyl, C(O)-alkyl, S(O).sub.2-alkyl and benzyl is optionally substituted, where chemically possible, by 1 to 4 substituents which

are each independently selected at each occurrence from the group consisting of: =O; =NR^{sup.a}, =NOR^{sup.a}, C_{sub.1}-C_{sub.4}-alkyl, halo, nitro, cyano, C_{sub.1}-C_{sub.4}-haloalkyl, C_{sub.2}-C_{sub.4}-alkenyl, C_{sub.2}-C_{sub.4}-alkynyl, NR^{sup.a}R^{sup.b}, S(O)_{sub.2}R^{sup.a}, S(O)R^{sup.a}, S(O)(NR^{sup.a})R^{sup.a}, S(O)_{sub.2}NR^{sup.a}R^{sup.a}, CO_{sub.2}R^{sup.a}, C(O)R^{sup.a}, CONR^{sup.a}R^{sup.a}, OR^{sup.a} and SR^{sup.a}; wherein R^{sup.a} is independently selected from H and C_{sub.1}-C_{sub.4}-alkyl; and R^{sup.b} is independently H, C_{sub.1}-C_{sub.4}-alkyl, C(O)—C_{sub.1}-C_{sub.4}-alkyl, S(O)_{sub.2}—C_{sub.1}-C_{sub.4}-alkyl.

(15) In certain embodiments, the compound of formula I is a compound of formula II:

(16) ##STR00002##

wherein Y^{sup.1}, Y^{sup.2}, R^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4}, m and y are as described above for formula I.

(17) In certain embodiments, the compound of formula I is a compound of formula III:

(18) ##STR00003##

wherein R^{sup.2}, R^{sup.3}, R^{sup.4} and m are as described above for formula I.

(19) In certain embodiments, the compound of formula I is a compound of formula IV:

(20) ##STR00004##

wherein Z^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4} and m are as described above for formula I.

(21) In certain embodiments, the compound of formula I is a compound of formula V:

(22) ##STR00005##

wherein Z^{sup.1}, R^{sup.2}, R^{sup.4}, R^{sup.12}, and m are as described above for formula I; and x is an integer selected from 0, 1, 2, 3, 4 and 5.

(23) In certain embodiments, the compound of formula I is a compound of formula VI:

(24) ##STR00006##

wherein R^{sup.2}, R^{sup.4}, R^{sup.12}, and m are as described above for formula I; and x is an integer selected from 0, 1, 2, 3, 4 and 5.

(25) The following embodiments apply to compounds of any of formulae (I)-(VI). These embodiments are independent and interchangeable. Any one embodiment may be combined with any other embodiment, where chemically allowed. In other words, any of the features described in the following embodiments may (where chemically allowable) be combined with the features described in one or more other embodiments. In particular, where a compound is exemplified or illustrated in this specification, any two or more of the embodiments listed below, expressed at any level of generality, which encompass that compound may be combined to provide a further embodiment which forms part of the present disclosure.

(26) It may be that R^{sup.1} is independently at each occurrence selected from C_{sub.1}-C_{sub.4}-alkyl, halo and OR^{sup.9}.

(27) R^{sup.4} may be independently selected from C_{sub.1}-C_{sub.4}-alkyl, C_{sub.3}-C_{sub.6}-cycloalkyl and C_{sub.1}-C_{sub.6}-haloalkyl. R^{sup.4} may be independently selected from C_{sub.1}-C_{sub.6}-alkyl and C_{sub.3}-C_{sub.6}-cycloalkyl. R^{sup.4} may be methyl or ethyl. R^{sup.4} may be methyl.

(28) y may be 0.

(29) Y^{sup.1} may be S. Y^{sup.1} may be O.

(30) =Y^{sup.2} may be S. =Y^{sup.2} may be O.

(31) It may be that X^{sup.1} and X^{sup.2} are each carbon. It may be that at least one of X^{sup.1} and X^{sup.2} is nitrogen. It may be that a single one of X^{sup.1} and X^{sup.2} is nitrogen. It may be that X^{sup.2} is nitrogen. It may be that X^{sup.2} is nitrogen and X^{sup.1} is carbon. It may be that X^{sup.1} is carbon and X^{sup.2} is independently selected from carbon and nitrogen.

(32) m may be 0. m may be 1.

(33) It may be that R^{sup.1}, R^{sup.2} and R^{sup.12} are each independently at each occurrence selected from C_{sub.1}-C_{sub.6}-alkyl, C_{sub.1}-C_{sub.6}-haloalkyl, C_{sub.3}-C_{sub.6}-cycloalkyl, halogen, nitro, OR^{sup.9}, SR^{sup.10}, OS(O)_{sub.2}R^{sup.10}, S(O)_{sub.2}R^{sup.10}, C(O)OR^{sup.10},

C(O)NR.sup.10R.sup.10, C(O)R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, and NR.sup.10R.sup.11. It may be that R.sup.2 is independently at each occurrence selected from C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.3-C.sub.6-cycloalkyl, halogen, nitro, OR.sup.9, SR.sup.10, OS(O).sub.2R.sup.10, S(O).sub.2R.sup.10, C(O)OR.sup.10, C(O)NR.sup.10R.sup.10, C(O)R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, and NR.sup.10R.sup.11.

(34) R.sup.2 may be independently at each occurrence selected from: cyano, nitro, C.sub.1-C.sub.4-alkyl, halo, O-R.sup.9 and phenyl optionally substituted with from 1 to 5 R.sup.12 groups. R.sup.2 may be independently at each occurrence selected from: cyano, nitro, C.sub.1-C.sub.4-alkyl, halo and O—R.sup.9.

(35) R.sup.2 may be independently at each occurrence phenyl optionally substituted with from 1 to 5 R.sup.12 groups.

(36) R.sup.2 may be phenyl substituted with a single R.sup.12. R.sup.2 may be phenyl substituted with R.sup.12, wherein R.sup.12 is halogen.

(37) R.sup.2 may be unsubstituted phenyl.

(38) Z.sup.1 may be absent or independently selected from C(O)O, OC(O), O, S, S(O), S(O).sub.2, C.sub.1-C.sub.3-alkylene and N(C.sub.1-C.sub.4-alkyl). Z.sup.1 may be absent or independently selected from C(O)O, OC(O), O, S, C.sub.1-C.sub.3-alkylene and N(C.sub.1-C.sub.4-alkyl).

Z.sup.1 may be independently selected from C.sub.1-C.sub.3-alkylene, S, N(C.sub.1-C.sub.4-alkyl) and O. Z.sup.1 may be independently selected from C.sub.1-alkylene, S, N(C.sub.1-C.sub.4-alkyl) and O. Z.sup.1 may be independently selected from S, N(C.sub.1-C.sub.4-alkyl) and O. Z.sup.1 may be independently selected from C.sub.1-alkylene, S, and O. Z.sup.1 may be absent or independently selected from C.sub.1-C.sub.3-alkylene and O. Z.sup.1 may be absent or it may be O. Z.sup.1 may be O. Z.sup.1 may be a group selected from S and O. Z.sup.1 may be absent.

Z.sup.1 may be C(O)O. Z.sup.1 may be S. Z.sup.1 may be C.sub.1-C.sub.3-alkylene. Z.sup.1 may be C.sub.1-alkylene. Z.sup.1 may be N(C.sub.1-C.sub.4-alkyl) e.g. N-Me. Illustrative, C.sub.1-alkylene groups that Z.sub.1 may be include CH.sub.2, CH(OH), C(NHOR.sup.a) and C(O).

(39) R.sup.3 may be CH.sub.2R.sup.3a. Alternatively, R.sup.3 may be R.sup.3a.

(40) R.sup.3a may be optionally substituted phenyl, e.g. unsubstituted phenyl. R.sup.3 may be optionally substituted phenyl, e.g. unsubstituted phenyl.

(41) R.sup.3 may have the structure:

(42) ##STR00007##

wherein x is an integer selected from 0, 1, 2, 3, 4 and 5.

(43) Z.sup.1-R.sup.3 may have the structure:

(44) ##STR00008##

wherein x is an integer selected from 0, 1, 2, 3, 4 and 5.

(45) x may be at least 1. x may be 0. x may be 1. x may be 2 or 3.

(46) R.sup.12 may be independently at each occurrence selected from: cyano, nitro, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, halo and S—R.sup.10, O—R.sup.9. R.sup.12 may be independently at each occurrence selected from: cyano, nitro, C.sub.1-C.sub.4-alkyl, halo and S—R.sup.10, O—R.sup.9. R.sup.12 may be independently at each occurrence selected from: C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl and halo. R.sup.12 may in one occurrence be situated para to Z.sup.1. R.sup.12 may be halo e.g. Chloro. R.sup.12 may in one occurrence be situated ortho to Z.sup.1. R.sup.12 may in one occurrence be situated meta to Z.sup.1.

(47) It may be that R.sup.9 is independently at each occurrence selected from: H, C.sub.1-C.sub.6-alkyl, C.sub.3-C.sub.6-cycloalkyl, C(O)—C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.6-haloalkyl. It may be that R.sup.9 is independently at each occurrence selected from: C.sub.1-C.sub.6-alkyl and C.sub.0-C.sub.3-alkylene-C.sub.3-C.sub.6-cycloalkyl. R.sup.3 may be optionally substituted heteroaryl, e.g. optionally substituted 6-membered heteroaryl. R.sup.3 may be optionally

substituted pyridine. R.sup.3 may be independently selected from optionally substituted phenyl and optionally substituted pyridine.

(48) R.sup.3 may be optionally substituted 5-, 6-7, or 8 membered heterocycloalkyl. R.sup.3 may be optionally substituted 6-membered heterocycloalkyl, e.g. piperidine or piperazine.

(49) R.sup.3 may be selected from C.sub.3-C.sub.6-alkyl and C.sub.0-C.sub.3-alkylene-R.sup.3a wherein R.sup.3a is C.sub.3-C.sub.3-cycloalkyl. R.sup.3 may be C.sub.3-C.sub.3-alkyl. R.sup.3 may be C.sub.3-C.sub.6-alkyl. R.sup.3 may be C.sub.5-C.sub.8-alkyl. R.sup.3 may be C.sub.4-alkyl. R.sup.3 may be selected from butyl, isobutyl, secbutyl and tertbutyl. R.sup.3 may be selected from isobutyl and secbutyl. R.sup.3 may be isobutyl. R.sup.3 may be C.sub.0-C.sub.3-alkylene-R.sup.3a wherein R.sup.3a is C.sub.3-C.sub.3-cycloalkyl. R.sup.3 may be C.sub.1-C.sub.3-alkylene-R.sup.3a wherein R.sup.3a is C.sub.3-C.sub.3-cycloalkyl. R.sup.3 may be CH.sub.2—R.sup.3a wherein R.sup.3a is C.sub.3-C.sub.3-cycloalkyl. R.sup.3 may be C.sub.3-C.sub.3-cycloalkyl. Said R.sup.3 groups may be unsubstituted. Said R.sub.3 groups may be substituted with from 1 to 4 groups selected from F and C.sub.1-C.sub.2-alkyl.

(50) R.sup.3 may have a structure selected from:

(51) ##STR00009##

(52) The compound of formula (I) may be selected from:

(53) ##STR00010## ##STR00011## ##STR00012## ##STR00013## ##STR00014##

##STR00015## ##STR00016## ##STR00017##

(54) The present invention is further defined by the following numbered clauses:

(55) 1. A compound of formula I, or an agronomically acceptable salt or N-oxide thereof:

(56) ##STR00018## X.sup.1 and X.sup.2 are each independently selected from carbon and nitrogen; Y.sup.1 is independently selected from O and S; =Y.sup.2 is independently selected from =O and =S Z.sup.1 is independently absent or is selected from C(O)O, OC(O), O, S, S(O), S(O).sub.2, C(O)NR.sup.5, NR.sup.5C(O), S(O).sub.2NR.sup.5, NR.sup.5S(O).sub.2, S(O)NR.sup.5, NR.sup.5S(O), CR.sup.6R.sup.7, C(O), C(S), C=NOR.sup.3, C.sub.1-C.sub.3-alkylene and NR.sup.5; R.sup.1, R.sup.2 and R.sup.12 are each independently at each occurrence selected from C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.3-C.sub.6-cycloalkyl, halogen, nitro, OR.sup.9, SR.sup.10, OS(O).sub.2R.sup.10, S(O).sub.2R.sup.10, C(O)OR.sup.10, C(O)NR.sup.10R.sup.10, C(O)R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, and NR.sup.10R.sup.11; R.sup.3 is independently at each occurrence selected from C.sub.3-C.sub.8-alkyl and C.sub.0-C.sub.3-alkylene-R.sup.3a; wherein R.sup.3a is independently at each occurrence selected from phenyl, 5- or 6-membered heteroaryl, 5-, 6-, 7- or 8-membered heterocycloalkyl and C.sub.3-C.sub.8-cycloalkyl; said heterocycloalkyl or cycloalkyl group being monocyclic or bicyclic; said heteroaryl or phenyl group being optionally substituted with from 1 to 5 R.sup.12 groups or said heterocycloalkyl or cycloalkyl group being optionally substituted with from 1 to 4 R.sup.13 groups; said heterocycloalkyl or cycloalkyl group being optionally fused to phenyl or a 5- or 6-membered heteroaryl, said heteroaryl or phenyl group being optionally substituted with from 1 to 4 R.sup.12 groups; R.sup.4 is independently at each occurrence selected from: C.sub.1-C.sub.6-alkyl, C.sub.3-C.sub.6-cycloalkyl and C.sub.1-C.sub.6-haloalkyl; R.sup.5 and R.sup.10 are each independently at each occurrence selected from: H, C.sub.3-C.sub.6-cycloalkyl, C.sub.1-C.sub.6-alkyl and benzyl; or where two R.sup.10 groups are attached to the same nitrogen atom, said R.sup.10 groups, together with said nitrogen atom form a 4-, 5-, 6- or 7-membered heterocycloalkyl ring; R.sup.6 is independently at each occurrence selected from: H, C.sub.3-C.sub.6-cycloalkyl, C.sub.1-C.sub.6-alkyl, phenyl and 5- or 6-membered heteroaryl; R.sup.7 is independently at each occurrence selected from: H, halo and OR.sup.8; R.sup.8 is each independently at each occurrence selected from: H, C.sub.3-C.sub.6-cycloalkyl, C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.3-alkylene-R.sup.8a; wherein R.sup.8a is independently at each occurrence selected from phenyl and 5- or 6-membered heteroaryl; R.sup.9 is independently at each occurrence selected from: H, C.sub.1-

C.sub.6-alkyl, C.sub.3-C.sub.6-cycloalkyl, C(O)—C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.6-haloalkyl; R.sup.11 is independently at each occurrence selected from; H, C.sub.1-C.sub.6-alkyl, C(O)—C.sub.1-C.sub.6-alkyl and S(O).sub.2—C.sub.1-C.sub.6-alkyl; or where an R.sup.10 group and an R.sup.11 group are attached to the same nitrogen atom, said R.sup.10 and R.sup.11 groups, together with said nitrogen atom form a 4-, 5-, 6- or 7-membered heterocycloalkyl ring; R.sup.13 is independently at each occurrence selected from: =O, =S, C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.3-C.sub.6-cycloalkyl; halogen, nitro, OR.sup.9, SR.sup.10, OS(O).sub.2R.sup.10, S(O).sub.2R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, and NR.sup.10R.sup.11; y is independently an integer selected from 0, 1 and 2; m is independently an integer selected from 0, 1, 2 and 3; and wherein any aforementioned alkyl, alkylene, alkenyl, cycloalkyl, heterocycloalkyl (including where two R.sup.10 groups or an R.sup.10 group and an R.sup.11 group together with a nitrogen to which they are attached form a heterocycloalkyl ring), alkynyl, C(O)-alkyl, S(O).sub.2-alkyl and benzyl is optionally substituted, where chemically possible, by 1 to 4 substituents which are each independently selected at each occurrence from the group consisting of: =O; =NR.sup.a, =NOR.sup.a, C.sub.1-C.sub.4-alkyl, halo, nitro, cyano, C.sub.1-C.sub.4-haloalkyl, C.sub.2-C.sub.4-alkenyl, C.sub.2-C.sub.4-alkynyl, NR.sup.aR.sup.b, S(O).sub.2R.sup.a, S(O)R.sup.a, S(O)(NR.sup.a)R.sup.a, S(O).sub.2NR.sup.aR.sup.a, CO.sub.2R.sup.a, C(O)R.sup.a, CONR.sup.aR.sup.a, OR.sup.a and SR.sup.a; wherein R.sup.a is independently selected from H and C.sub.1-C.sub.4-alkyl; and R.sup.b is independently H, C.sub.1-C.sub.4-alkyl, C(O)—C.sub.1-C.sub.4-alkyl, S(O).sub.2—C.sub.1-C.sub.4-alkyl.

(57) 2. A compound of clause 1, wherein R.sup.4 is independently selected from: C.sub.1-C.sub.6-alkyl and C.sub.3-C.sub.6-cycloalkyl.

(58) 3. A compound of clause 1 or clause 2, wherein y is 0.

(59) 4. A compound of any one of clauses 1 to 3, wherein Y.sup.1 is O.

(60) 5. A compound of any one of clauses 1 to 4, wherein =Y.sup.2 is =S.

(61) 6. A compound of any one of clauses 1 to 5, wherein X.sup.1 is carbon.

(62) 7. A compound of clause 6, wherein X.sup.2 is carbon.

(63) 8. A compound of clause 6, wherein X.sup.2 is nitrogen.

(64) 9. A compound of any one of clauses 1 to 8, wherein Z.sup.1 is independently at each occurrence selected from O, S, C.sub.1-alkylene and NR.sup.5, wherein R.sup.5 is C.sub.1-C.sub.3-alkyl

(65) 10. A compound of clause 9, wherein Z.sup.1 is O.

(66) 11. A compound of any one of clauses 1 to 10, wherein R.sup.3 is R.sup.3a.

(67) 12. A compound of any one of clauses 1 to 11, wherein R.sup.3a is optionally substituted phenyl.

(68) 13. A compound of any one of clauses 1 to 10, wherein R.sup.3 is C.sub.3-C.sub.3-alkyl.

(69) 14. A compound of any one of clauses 1 to 13, wherein m is 0.

(70) 15. A compound of clause 1, wherein the compound of formula (I) is selected from:

(71) ##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023##
##STR00024## ##STR00025## ##STR00026## ##STR00027##

(72) 16. A method for controlling fungal diseases, the method comprising applying an agronomically effective and substantially non-phytotoxic quantity of a compound of any one of clauses 1 to 15 to seeds of plants, to plants themselves or to an area where it is intended that plants will grow.

(73) 17. A use of a compound of any one of clauses 1 to 15 to control fungal diseases of plants.

(74) 18. A fungicidal composition comprising an effective and non-phytotoxic amount of an active compound of any one of clauses 1 to 15.

Description

DETAILED DESCRIPTION

(1) The term C.sub.m-C.sub.n refers to a group with m to n carbon atoms.

(2) The term “alkyl” refers to a linear or branched saturated monovalent hydrocarbon chain. For example, C.sub.1-C.sub.6-alkyl may refer to methyl, ethyl, n-propyl, iso-propyl, n-butyl, sec-butyl, tert-butyl, n-pentyl and n-hexyl. The alkyl groups may be unsubstituted or substituted by one or more substituents. Specific substituents for each alkyl group independently may be fluorine, OR.sup.a or NHR.sup.a.

(3) The term “alkylene” refers to a linear saturated divalent hydrocarbon chain. The alkylene groups may be unsubstituted or substituted by one or more substituents. Specific substituents for each alkylene group independently may be C.sub.1-C.sub.4-alkyl, fluorine, OR.sup.a or NHR.sup.a.

(4) The term “haloalkyl” refers to a hydrocarbon group substituted with at least one halogen atom independently chosen at each occurrence from: fluorine, chlorine, bromine and iodine. The halogen atom may be present at any position on the hydrocarbon chain. For example, C.sub.1-C.sub.6-haloalkyl may refer to chloromethyl, fluoromethyl, trifluoromethyl, chloroethyl e.g. 1-chloroethyl and 2-chloroethyl, trichloroethyl e.g. 1,2,2-trichloroethyl, 2,2,2-trichloroethyl, fluoroethyl e.g. 1-fluoroethyl and 2-fluoroethyl, trifluoroethyl e.g. 1,2,2-trifluoroethyl and 2,2,2-trifluoroethyl, chloropropyl, trichloropropyl, fluoropropyl, trifluoropropyl. A haloalkyl group may be a fluoroalkyl group, i.e. a hydrocarbon chain substituted with at least one fluorine atom. Thus, a haloalkyl group may have any amount of halogen substituents. The group may contain a single halogen substituent, it may have two or three halogen substituents, or it may be saturated with halogen substituents.

(5) The term “alkenyl” refers to a branched or linear hydrocarbon group containing at least one double bond. The double bond(s) may be present as the E or Z isomer. The double bond may be at any possible position of the hydrocarbon chain; for example, “C.sub.2-C.sub.6-alkenyl” may refer to ethenyl, propenyl, butenyl, butadienyl, pentenyl, pentadienyl, hexenyl and hexadienyl. The alkenyl groups may be unsubstituted or substituted by one or more substituents. Specific substituents for any saturated carbon atom in each alkenyl group independently may be fluorine, OR.sup.a or NHR.sup.a.

(6) The term “alkynyl” refers to a branched or linear hydrocarbon chain containing at least one triple bond. The triple bond may be at any possible position of the hydrocarbon chain. For example, “C.sub.2-C.sub.6-alkynyl” may refer to ethynyl, propynyl, butynyl, pentynyl and hexynyl. The alkynyl groups may be unsubstituted or substituted by one or more substituents. Specific substituents for any saturated carbon atom in each alkynyl group independently may be fluorine, OR.sup.a or NHR.sup.a.

(7) The term “cycloalkyl” refers to a saturated hydrocarbon ring system containing, for example, 3, 4, 5 or 6 carbon atoms. For example, “C.sub.3-C.sub.6-cycloalkyl” may refer to cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl. The cycloalkyl groups may be unsubstituted or substituted by one or more substituents. Specific substituents for each cycloalkyl group independently may be fluorine, OR.sup.a or NHR.sup.a.

(8) The term heterocycloalkyl may refer to a monocyclic or bicyclic saturated or partially saturated group having the indicated number of atoms in the ring system and comprising 1 or 2 heteroatoms independently selected from O, S and N in the ring system (in other words 1 or 2 of the atoms forming the ring system are selected from O, S and N). By partially saturated it is meant that the ring may comprise one or two double bonds. This applies particularly to monocyclic rings with from 5 to 6 members. The double bond will typically be between two carbon atoms but may be between a carbon atom and a nitrogen atom. Examples of heterocycloalkyl groups include;

piperidine, piperazine, morpholine, thiomorpholine, pyrrolidine, tetrahydrofuran, tetrahydrothiophene, dihydrofuran, tetrahydropyran, dihydropyran, dioxane, azepine. A heterocycloalkyl group may be unsubstituted or substituted by one or more substituents. Specific substituents for any saturated carbon atom in each heterocycloalkyl group may independently be fluorine, OR^{sup.a} or NHR^{sup.a}.

(9) Aryl groups may be any aromatic carbocyclic ring system (i.e. a ring system containing $2(2n+1)\pi$ electrons). Aryl groups may have from 6 to 12 carbon atoms in the ring system. Aryl groups will typically be phenyl groups. Aryl groups may be naphthyl groups or biphenyl groups.

(10) In any of the above aspects and embodiments, heteroaryl groups may be any aromatic (i.e. a ring system containing $2(2n+1)\pi$ electrons) 5-10 membered ring system comprising from 1 to 4 heteroatoms independently selected from O, S and N (in other words from 1 to 4 of the atoms forming the ring system are selected from O, S and N). Thus, any heteroaryl groups may be independently selected from: 5 membered heteroaryl groups in which the heteroaromatic ring is substituted with 1-4 heteroatoms independently selected from O, S and N; and 6-membered heteroaryl groups in which the heteroaromatic ring is substituted with 1-3 (e.g. 1-2) nitrogen atoms; 9-membered bicyclic heteroaryl groups in which the heteroaromatic system is substituted with 1-4 heteroatoms independently selected from O, S and N; 10-membered bicyclic heteroaryl groups in which the heteroaromatic system is substituted with 1-4 nitrogen atoms. Specifically, heteroaryl groups may be independently selected from: pyrrole, furan, thiophene, pyrazole, imidazole, oxazole, isoxazole, triazole, oxadiazole, thiadiazole, tetrazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, indole, isoindole, benzofuran, isobenzofuran, benzothiophene, indazole, benzimidazole, benzoxazole, benzothiazole, benzisoxazole, purine, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, pteridine, phthalazine, naphthyridine.

(11) It may be that, in any group which is an aryl or heteroaryl group, that aryl or heteroaryl group is unsubstituted or is optionally substituted, where chemically possible, by 1 to 5 substituents which are each independently selected at each occurrence from: halo, nitro, cyano,

NR^{sup.a}, R^{sup.a}, NR^{sup.a}S(O)₂R^{sup.a}, NR^{sup.a}C(O)R^{sup.a}, NR^{sup.a}CONR^{sup.a}R^{sup.a}, NR^{sup.a}CO₂R^{sup.a}, OR^{sup.a}, SR^{sup.a}, S(O)₂R^{sup.a}, S(O)₂OR^{sup.a}, S(O)₂R^{sup.a}, S(O)₂NR^{sup.a}R^{sup.a}, CO₂R^{sup.a}, C(O)R^{sup.a}, CONR^{sup.a}R^{sup.a}, CR^{sup.b}R^{sup.b}NR^{sup.a}R^{sup.a}, CR^{sup.b}R^{sup.b}OR^{sup.a}, C₁₋₄-alkyl, C₂₋₄-alkenyl, C₂₋₄-alkynyl and C₁₋₄-haloalkyl; wherein R^{sup.a} and R^{sup.b} are as described above for formula I.

(12) Compounds of the invention containing one or more asymmetric carbon atoms can exist as two or more stereoisomers. Where a compound of the invention contains a double bond such as a C=C or C=N group, geometric cis/trans (or Z/E) isomers are possible. Where structural isomers are interconvertible via a low energy barrier, tautomeric isomerism ('tautomerism') can occur. This can take the form of proton tautomerism in compounds of the invention containing, for example, an imino, keto, or oxime group, or so-called valence tautomerism in compounds which contain an aromatic moiety. It follows that a single compound may exhibit more than one type of isomerism.

(13) Included within the scope of the present invention are all stereoisomers, geometric isomers and tautomeric forms of the compounds of the invention, including compounds exhibiting more than one type of isomerism, and mixtures of one or more thereof.

(14) The compounds of the invention may be obtained, stored and/or used in the form of an agronomically acceptable salt. Suitable salts include, but are not limited to, salts of acceptable inorganic acids such as hydrochloric, sulfuric, phosphoric, nitric, carbonic, boric, sulfamic, and hydrobromic acids, or salts of agronomically acceptable organic acids such as acetic, propionic, butyric, tartaric, maleic, hydroxymaleic, fumaric, malic, citric, lactic, mucic, gluconic, benzoic, succinic, oxalic, phenylacetic, methanesulfonic, toluenesulfonic, benzenesulfonic, salicylic, sulfanilic, aspartic, glutamic, edetic, stearic, palmitic, oleic, lauric, pantothenic, tannic, ascorbic and valeric acids. Suitable salts also include salts of inorganic and organic bases, e.g. counterions

such as Na, Ca, K, Li, Mg, ammonium, trimethylsulfonium. The compounds may also be obtained, stored and/or used in the form of an N-oxide. Also included are acid addition salts or base salts wherein the counter ion is optically active; for example, d-lactate or l-lysine, or racemic; for example, dl-tartrate or dl-arginine.

(15) Cis/trans isomers may be separated by conventional techniques well known to those skilled in the art, for example, chromatography and fractional crystallisation.

(16) Conventional techniques for the preparation/isolation of individual enantiomers when necessary include chiral synthesis from a suitable optically pure precursor or resolution of the racemate (or the racemate of a salt or derivative) using, for example, chiral high pressure liquid chromatography (HPLC). Thus, chiral compounds of the invention (and chiral precursors thereof) may be obtained in enantiomerically-enriched form using chromatography, typically HPLC, on an asymmetric resin with a mobile phase consisting of a hydrocarbon, typically heptane or hexane, containing from 0 to 50% by volume of isopropanol, typically from 2% to 20%, and for specific examples, 0 to 5% by volume of an alkylamine e.g. 0.1% diethylamine. Concentration of the eluate affords the enriched mixture.

(17) Alternatively, the racemate (or a racemic precursor) may be reacted with a suitable optically active compound, for example, an alcohol, or, in the case where the compound of the invention contains an acidic or basic moiety, a base or acid such as 1-phenylethylamine or tartaric acid. The resulting diastereomeric mixture may be separated by chromatography and/or fractional crystallisation and one or both of the diastereoisomers converted to the corresponding pure enantiomer(s) by means well known to a skilled person.

(18) When any racemate crystallises, crystals of two different types are possible. The first type is the racemic compound (true racemate) referred to above wherein one homogeneous form of crystal is produced containing both enantiomers in equimolar amounts. The second type is the racemic mixture or conglomerate wherein two forms of crystal are produced in equimolar amounts each comprising a single enantiomer.

(19) While both of the crystal forms present in a racemic mixture have identical physical properties, they may have different physical properties compared to the true racemate. Racemic mixtures may be separated by conventional techniques known to those skilled in the art—see for example, “Stereochemistry of Organic Compounds” by E. L. Eliel and S. H. Wilen (Wiley, 1994).

(20) The activity of the compounds of the present invention can be assessed by a variety of *in silico*, *in vitro* and *in vivo* assays. *In silico* analysis of a variety of compounds has been demonstrated to be predictive of ultimate *in vitro* and even *in vivo* activity.

(21) The present invention also includes all environmentally acceptable isotopically-labelled compounds of formulae I to VI and their syntheses, wherein one or more atoms are replaced by atoms having the same atomic number, but an atomic mass or mass number different from the atomic mass or mass number usually found in nature.

(22) Examples of isotopes suitable for inclusion in the compounds of the invention include isotopes of hydrogen, such as ²H and ³H, carbon, such as ¹¹C, ¹³C and ¹⁴C, chlorine, such as ³⁶Cl, fluorine, such as ¹⁸F, iodine, such as ¹²³I and ¹²⁵I, nitrogen, such as ¹³N and ¹⁵N, oxygen, such as ¹⁵O, ¹⁷O and ¹⁸O, phosphorus, such as ³²P, and sulfur, such as ³⁵S.

(23) Isotopically-labelled compounds can generally be prepared by conventional techniques known to those skilled in the art or by processes analogous to those described using an appropriate isotopically-labelled reagent in place of the non-labelled reagent previously employed.

(24) Throughout the description and claims of this specification, the words “comprise” and “contain” and variations of the words, for example “comprising” and “comprises”, means “including but not limited to”, and is not intended to (and does not) exclude other moieties, additives, components, integers or steps.

(25) Throughout the description and claims of this specification, the singular encompasses the

plural unless the context otherwise requires. In particular, where the indefinite article is used, the specification is to be understood as contemplating plurality as well as singularity, unless the context requires otherwise.

(26) Features, integers, characteristics, compounds, chemical moieties or groups described in conjunction with a particular aspect, embodiment or example of the invention are to be understood to be applicable to any other aspect, embodiment or example described herein unless incompatible therewith.

(27) If appropriate, the compounds of the invention can, at certain concentrations or application rates, be used as fungicides.

(28) According to another aspect of the present invention, there is provided a method for controlling fungal diseases, the method comprising the application of an agronomically effective and substantially non-phytotoxic (to the crop plant) quantity of a compound of the invention to the seeds of the plants, to the plants themselves or to the area where it is intended that the plants will grow.

(29) The pesticide may be applied as a seed treatment, foliar application, stem application, drench or drip application (chemigation) to the seed, the plant or to the fruit of the plant or to soil or to inert substrate (e.g. inorganic substrates like sand, rockwool, glasswool; expanded minerals like perlite, vermiculite, zeolite or expanded clay), Pumbe, Pyroclastic materials or stuff, synthetic organic substrates (e.g. polyurethane) organic substrates (e.g. peat, composts, tree waste products like coir, wood fibre or chips, tree bark) or to a liquid substrate (e.g. floating hydroponic systems, Nutrient Film Technique, Aeroponics).

(30) In a further aspect, the present invention also relates to a fungicidal composition comprising an effective and non-phytotoxic amount of an active compound of the invention. The composition may further comprise one or more additional fungicides.

(31) The term “effective and non-phytotoxic amount” means an amount of pesticide according to the invention which is sufficient to control or destroy any of the targeted pests present or liable to appear in the crops and which does not have any significant detrimental effect on the crops or indeed has a positive effect on plant vigour and yield in the absence of target organism. The amount will vary depending on the pest to be controlled, the type of crop, the climatic conditions and the compounds included in the pesticidal composition. This amount can be determined by systematic field trials, which are within the capabilities of a person skilled in the art.

(32) Depending on their particular physical and/or chemical properties, the active compounds of the invention can be formulated as solutions, emulsions, suspensions, powders, foams, pastes, granules, aerosols, microencapsulations in polymeric substances and in coating materials for seed, and also as ULV cold and warm fogging formulations.

(33) The active compounds can be used neat, or in the form of a formulation, e.g. ready-to-use solutions, emulsions, water- or oil-based suspensions, powders, wettable powders, pastes, soluble powders, dusts, soluble granules, granules for broadcasting, suspoemulsion concentrates, natural substances impregnated with active compound, synthetic substances impregnated with active compound, fertilizers and also microencapsulations in polymeric substances. Application may be carried out, for example, by watering, spraying, atomising, broadcasting, dusting, foaming, spreading, etc. It is also possible to apply the active compounds by the ultra-low volume method or to inject the preparation of active compound or the active compound itself into the soil. It is also possible to treat the seed of the plants.

(34) Formulations containing the compounds of the invention are produced in a known manner, for example by mixing the compounds with extenders (e.g. liquid solvents and/or solid carriers), optionally with the use of surfactants (e.g. emulsifiers and/or dispersants and/or foam-formers). The formulations are prepared either in factories/production plants or alternatively before or during the application.

(35) Auxiliaries are substances which are suitable for imparting to the composition itself and/or to

preparations derived therefrom (for example; spray liquors, seed dressings) particular properties such as certain technical properties and/or also particular biological properties. Typical suitable auxiliaries are: extenders, solvents and carriers.

(36) Suitable extenders are, for example, water, polar and nonpolar organic chemical liquids, for example from the classes of the aromatic and non-aromatic hydrocarbons (such as paraffins, alkylbenzenes, alkyl-naphthalenes, chlorobenzenes), the alcohols and polyols (which, if appropriate, may also be substituted, etherified and/or esterified), the ketones (such as acetone, cyclohexanone), esters (including fats and oils) and (poly)ethers, the unsubstituted and substituted amines, amides, lactams (such as N-alkylpyrrolidones) and lactones, the sulfones and sulfoxides (such as dimethyl sulfoxide).

(37) If the extender used is water, it is also possible to use, for example, organic solvents as auxiliary solvents. Essentially, suitable liquid solvents are: aromatics such as xylene, toluene or alkyl-naphthalenes; chlorinated aromatics and chlorinated aliphatic hydrocarbons such as chlorobenzenes, chloroethylenes or methylene chloride; aliphatic hydrocarbons such as cyclohexane or paraffins, for example petroleum fractions; alcohols such as butanol or glycol and also their ethers and esters; ketones such as acetone, methyl ethyl ketone, methyl isobutyl ketone or cyclohexanone; strongly polar solvents such as dimethylformamide and dimethyl sulfoxide.

(38) Suitable solid carriers are: for example, ammonium salts and ground natural minerals such as kaolins, clays, talc, chalk, quartz, attapulgite, montmorillonite or diatomaceous earth, and ground synthetic minerals, such as finely divided silica, alumina and silicates; suitable solid carriers for granules are: for example, crushed and fractionated natural rocks such as calcite, marble, pumice, sepiolite and dolomite, and also synthetic granules of inorganic and organic meals, and granules of organic material such as paper, sawdust, coconut shells, maize cobs and tobacco stalks; suitable emulsifiers and/or foam-formers are: for example, nonionic and anionic emulsifiers, such as polyoxyethylene fatty acid esters, polyoxyethylene fatty alcohol ethers, for example alkylaryl polyglycol ethers, alkylsulfonates, alkyl sulfates, arylsulfonates and also protein hydrolysates; suitable dispersants are nonionic and/or ionic substances, for example from the classes of the alcohol-POE and/or -POP ethers, acid and/or POP-POE esters, alkylaryl and/or POP-POE ethers, fat- and/or POP-POE adducts, POE- and/or POP-polyol derivatives, POE- and/or POP-sorbitan- or -sugar adducts, alkyl or aryl sulfates, alkyl- or arylsulfonates and alkyl or aryl phosphates or the corresponding PO-ether adducts. Furthermore, suitable oligo- or polymers, for example those derived from vinylic monomers, from acrylic acid, from EO and/or PO alone or in combination with, for example, (poly)alcohols or (poly)amines. It is also possible to employ lignin and its sulfonic acid derivatives, unmodified and modified celluloses, aromatic and/or aliphatic sulfonic acids and their adducts with formaldehyde.

(39) Tackifiers such as carboxymethylcellulose and natural and synthetic polymers in the form of powders, granules or lattices, such as gum arabic, polyvinyl alcohol and polyvinyl acetate, as well as natural phospholipids such as cephalins and lecithins, and synthetic phospholipids, can be used in the formulations.

(40) Further additives may be mineral and vegetable oils. It is also possible to add colourants such as inorganic pigments, for example iron oxide, titanium oxide and Prussian Blue, and organic dyestuffs, such as alizarin dyestuffs, azo dyestuffs and metal phthalocyanine dyestuffs, and trace nutrients such as salts of iron, manganese, boron, copper, cobalt, molybdenum and zinc. Other possible additives are perfumes, mineral or vegetable, optionally modified oils and waxes.

(41) The formulations may also comprise stabilizers, e.g. low-temperature stabilizers, preservatives, antioxidants, light stabilizers or other agents which improve chemical and/or physical stability.

(42) The formulations generally comprise between 0.01 and 98% by weight of active compound, preferably between 0.1 and 95% and particularly preferably between 0.5 and 90%.

(43) The active compounds of the invention can also be used as a mixture with other known

fungicides, for example, to improve the activity spectrum or to reduce or slow the development of resistance. A mixture with other known active compounds such as nematocides, herbicides, insecticides, acaricides, or bactericides, or with fertilizers and growth regulators, safeners or semiochemicals is also possible.

(44) Exemplary application rates of the active compounds according to the invention are: when treating leaves: from 0.1 to 10 000 g/ha, preferably from 10 to 1000 g/ha, particularly preferably from 50 to 300 g/ha (when the application is carried out by watering or dripping, it is even possible to reduce the application rate, especially when inert substrates such as rock wool or perlite are used); when treating seed: from 2 to 200 g per 100 kg of seed, preferably from 2.5 to 150 g per 100 kg of seed, and particularly preferably from 2.5 to 25 g per 100 kg of seed, very particularly preferably from 2.5 to 12.5 g per 100 kg of seed; when treating the soil: from 0.1 to 10000 g/ha, preferably from 1 to 5000 g/ha.

(45) The compositions according to the invention are suitable for protecting any plant variety which is employed in agriculture, in the greenhouse, in forests or in horticulture and, in particular, cereals (e.g. wheat, barley, rye, millet and oats), maize, cotton, soya beans, rice, potatoes, sunflowers, beans, coffee, beet (for example sugar beet and fodder beet), peanuts, vegetables (e.g. tomatoes, cucumbers, onions and lettuce), lawns, fruit and nut trees (e.g. apples, pears, peaches, nectarines, apricots, hazelnut, pecan, macadamia, pistachio), soft fruit (e.g. strawberries, raspberries, blackcurrants, redcurrants), grapevines, bananas, cocoa and ornamental plants.

(46) The active compounds of the invention, in combination with good plant tolerance and favourable toxicity to warm-blooded animals and being tolerated well by the environment, are suitable for protecting plants and plant organs, for increasing the harvest yields, for improving the quality of the harvested material and for controlling pests, in particular fungal diseases, which are encountered in agriculture, in horticulture, in animal husbandry, in forests, in gardens and leisure facilities, in the protection of stored products and of materials, and in the hygiene sector. They may be preferably employed as crop protection agents.

(47) Use as Fungicides

(48) The compounds of the invention have activity as fungicides.

(49) The following are illustrative examples of agricultural pests that may be controlled by fungicidal compounds: Powdery mildew diseases such as: *Blumeria* diseases, caused for example by *Blumeria graminis*; *Podosphaera* diseases, caused for example by *Podosphaera leucotheca*; *Sphaerotheca* diseases, caused for example by *Sphaerotheca fuliginea*; *Uncinula* diseases, caused for example by *Uncinula necator*; Rust diseases such as: *Gymnosporangium* diseases, caused for example by *Gymnosporangium sabinae*; *Hemileia* diseases, caused for example by *Hemileia vastatrix*; *Phakopsora* diseases, caused for example by *Phakopsora pachyrhizi* or *Phakopsora meibomia*; *Puccinia* diseases, caused for example by *Puccinia recondita*; *Uromyces* diseases, caused for example by *Uromyces appendiculatus*; Oomycete diseases such as: Albugo diseases caused for example by *Albugo Candida*; *Bremia* diseases, caused for example by *Bremia lactucae*; *Peronospora* diseases, caused for example by *Peronospora pisi* or *P. brassicae*; *Phytophthora* diseases, caused for example by *Phytophthora infestans*; Plasmopara diseases, caused for example by *Plasmopara viticola*; *Pseudoperonospora* diseases, caused for example by *Pseudoperonospora humuli* or *Pseudoperonospora cubensis*; *Pythium* diseases, caused for example by *Pythium ultimum*; Leafspot, leaf blotch and leaf blight diseases such as: *Alternaria* diseases, caused for example by *Alternaria solani*; *Cercospora* diseases, caused for example by *Cercospora beticola*; *Cladosporium* diseases, caused for example by *Cladosporium cucumerinum*; *Cochliobolus* diseases, caused for example by *Cochliobolus sativus*; *Colletotrichum* diseases, caused for example by *Colletotrichum lindemuthianum*; *Cycloconium* diseases, caused for example by *Cycloconium oleaginum*; *Diaporthe* diseases, caused for example by *Diaporthe citri*; *Drechslera*, Syn: *Helminthosporium*) or *Cochliobolus miyabeanus*; *Elsinoe* diseases, caused for example by *Elsinoe fawcettii*; *Gloeosporium* diseases, caused for example by *Gloeosporium laeticolor*; *Glomerella*

diseases, caused for example by *Glomerella cingulata*; *Guignardia* diseases, caused for example by *Guignardia bidwelli*; *Leptosphaeria* diseases, caused for example by *Leptosphaeria maculans*; *Leptosphaeria nodorum*; *Magnaporthe* diseases, caused for example by *Magnaporthe grisea*; *Mycosphaerella* diseases, caused for example by *Mycosphaerella graminicola*; *Mycosphaerella arachidicola*; *Mycosphaerella fibensis*; *Phaeosphaeria* diseases, caused for example by *Phaeosphaeria nodorum*; *Pyrenophora* diseases, caused for example by *Pyrenophora teres*; *Ramularia* diseases, caused for example by *Ramularia collo-cygni*; *Rhynchosporium* diseases, caused for example by *Rhynchosporium secalis*; *Septoria* diseases, caused for example by *Septoria apii* or *Septoria lycopersici*; *Typhula* diseases, caused for example by *Typhula incarnata*; *Venturia* diseases, caused for example by *Venturia inaequalis*; Root and stem diseases such as: *Corticium* diseases, caused for example by *Corticium graminearum*; *Fusarium* diseases, caused for example by *Fusarium oxysporum*; *Gaeumannomyces* diseases, caused for example by *Gaeumannomyces graminis*; *Rhizoctonia* diseases, caused for example by *Rhizoctonia solani*; *Sarocladium* diseases caused for example by *Sarocladium oryzae*; *Sclerotium* diseases caused for example by *Sclerotium oryzae*; *Tapesia* diseases, caused for example by *Tapesia acuformis*; *Thielaviopsis* diseases, caused for example by *Thielaviopsis basicola*; Ear and panicle diseases including maize cob, such as: *Alternaria* diseases, caused for example by *Alternaria* spp.; *Aspergillus* diseases, caused for example by *Aspergillus flavus*; *Cladosporium* diseases, caused for example by *Cladosporium* spp.; *Claviceps* diseases, caused for example by *Claviceps purpurea*; *Fusarium* diseases, caused for example by *Fusarium culmorum*; *Gibberella* diseases, caused for example by *Gibberella zeae*; *Monographella* diseases, caused for example by *Monographella nivalis*; Smut and bunt diseases such as: *Sphacelotheca* diseases, caused for example by *Sphacelotheca reiliana*; *Tilletia* diseases, caused for example by *Tilletia caries*; *Urocystis* diseases, caused for example by *Urocystis occulta*; *Ustilago* diseases, caused for example by *Ustilago nuda*; Fruit rot and mould diseases such as: *Aspergillus* diseases, caused for example by *Aspergillus flavus*; *Botrytis* diseases, caused for example by *Botrytis cinerea*; *Penicillium* diseases, caused for example by *Penicillium expansum*; *Rhizopus* diseases caused for example by *Rhizopus stolonifer*; *Sclerotinia* diseases, caused for example by *Sclerotinia sclerotiorum*; *Verticillium* diseases, caused for example by *Verticillium albo-atrum*; Seed and soil borne decay, mould, wilt, rot and damping-off diseases such as: *Alternaria* diseases, caused for example by *Alternaria brassicicola*; *Aphanomyces* diseases, caused for example by *Aphanomyces euteiches*; *Ascochyta* diseases, caused for example by *Ascochyta lentis*; *Aspergillus* diseases, caused for example by *Aspergillus flavus*; *Cladosporium* diseases, caused for example by *Cladosporium herbarum*; *Cochliobolus* diseases, caused for example by *Cochliobolus sativus* (Conidiaform: *Drechslera*, *Bipolaris* Syn: *Helminthosporium*); *Colletotrichum* diseases, caused for example by *Colletotrichum coccodes*; *Fusarium* diseases, caused for example by *Fusarium culmorum*; *Gibberella* diseases, caused for example by *Gibberella zeae*; *Macrophomina* diseases, caused for example by *Macrophomina phaseolina*; *Monographella* diseases, caused for example by *Monographella nivalis*; *Penicillium* diseases, caused for example by *Penicillium expansum*; *Phoma* diseases, caused for example by *Phoma lingam*; *Phomopsis* diseases, caused for example by *Phomopsis sojae*; *Phytophthora* diseases, caused for example by *Phytophthora cactorum*; *Pyrenophora* diseases, caused for example by *Pyrenophora graminea*; *Pyricularia* diseases, caused for example by *Pyricularia oryzae*; *Pythium* diseases, caused for example by *Pythium ultimum*; *Rhizoctonia* diseases, caused for example by *Rhizoctonia solani*; *Rhizopus* diseases, caused for example by *Rhizopus oryzae*; *Sclerotium* diseases, caused for example by *Sclerotium rolfsii*; *Septoria* diseases, caused for example by *Septoria nodorum*; *Typhula* diseases, caused for example by *Typhula incarnata*; *Verticillium* diseases, caused for example by *Verticillium dahliae*; Canker, broom and dieback diseases such as: *Nectria* diseases, caused for example by *Nectria galligena*; Blight diseases such as: *Monilinia* diseases, caused for example by *Monilinia laxa*; Leaf blister or leaf curl diseases such as: *Exobasidium* diseases caused for example by *Exobasidium vexans*; *Taphrina* diseases, caused for

example by *Taphrina deformans*; Decline diseases of wooden plants such as: *Esca* diseases, caused for example by *Phaemoniella clamydospora*, *Phaeomoniella clamydospora*, *Phaeoacremonium aleophilum* and *Fomitiporia mediterranea*; *Eutypa* dyebark, caused for example by *Eutypa lata*; Dutch elm disease, caused for example by *Ceratocystis ulmi*; *Ganoderma* diseases caused for example by *Ganoderma boninense*; Diseases of flowers and seeds such as: *Botrytis* diseases, caused for example by *Botrytis cinerea*; Diseases of tubers such as: *Rhizoctonia* diseases, caused for example by *Rhizoctonia solani* *Helminthosporium* diseases, caused for example by *Helminthosporium solani*. Diseases of Tubers such as: *Rhizoctonia* diseases caused for example by *Rhizoctonia solani*; *Helminthosporium* diseases caused for example by *Helminthosporium solani*; Club root diseases such as: *Plasmodiophora* diseases, caused for example by *Plasmodiophora brassicae*.

(50) The compounds of the invention may be active against a broad spectrum of fungal diseases of plants. Alternatively, they may be active specifically against certain specific fungal diseases.

(51) Particular fungal diseases against which the compounds of the invention may be useful include: wheat leaf blotch (*Septoria tritici*), wheat brown rust (*Puccinia triticina*), wheat stripe rust (*Puccinia striiformis*), scab of apple (*Venturia inaequalis*), powdery mildew of grapevine (*Uncinula necator*), barley scald (*Rhynchosporium secalis*), blast of rice (*Magnaporthe grisea*), rust of soybean (*Phakopsora pachyrhizi*), glume blotch of wheat (*Leptosphaeria nodorum*), powdery mildew of wheat (*Blumeria graminis* f. sp. *tritici*), powdery mildew of barley (*Blumeria graminis* f. sp. *hordei*), powdery mildew of cucurbits (*Erysiphe dehor acearum*), anthracnose of cucurbits (*Glomerella lagenarium*), leaf spot of beet (*Cercospora beticola*), early blight of tomato (*Alternaria solani*), and spot blotch of barley (*Cochliobolus sativus*).

(52) In addition to their fungicidal activity, the compounds of the invention may also have activity against other microbes, e.g. bacteria.

(53) The fungicidal compounds of the invention may also be used in the treatment of fungal diseases of humans and animals (e.g. mammals). Likewise, the bactericidal compounds of the invention may be used in the treatment of bacterial diseases of humans and animals. Thus, the invention includes a method of treating a fungal or bacterial disease, the method comprising of administering a therapeutic amount of an antifungal agent of the invention to a subject (e.g. a human subject) in need thereof. The compound may be formulated for topical administration to the infected area of the body or it may be formulated for oral or parenteral administration.

(54) Synthesis

(55) The skilled person will appreciate that adaptation of methods known in the art could be applied in the manufacture of the compounds of the present invention.

(56) For example, the skilled person will be immediately familiar with standard textbooks such as "Comprehensive Organic Transformations—A Guide to Functional Group Transformations", RC Larock, Wiley-VCH (1999 or later editions); "March's Advanced Organic Chemistry—Reactions, Mechanisms and Structure", MB Smith, J. March, Wiley, (5th edition or later editions); "Advanced Organic Chemistry, Part B, Reactions and Synthesis", F A Carey, R J Sundberg, Kluwer Academic/Plenum Publications, (2001 or later editions); "Organic Synthesis—The Disconnection Approach", S Warren (Wiley), (1982 or later editions); "Designing Organic Syntheses" S Warren (Wiley) (1983 or later editions); "Heterocyclic Chemistry", J. Joule (Wiley 2010 edition or later editions); ("Guidebook To Organic Synthesis" R K Mackie and D M Smith (Longman) (1982 or later editions), etc., and the references therein as a guide.

(57) The skilled person is familiar with a range of strategies for synthesising organic and particularly heterocyclic molecules and these represent common general knowledge as set out in text books such as Warren "Organic Synthesis: The Disconnection Approach"; Mackie and Smith "Guidebook to Organic Chemistry"; and Clayden, Greeves, Warren and Wothers "Organic Chemistry".

(58) The skilled person will exercise his/her judgement and skill as to the most efficient sequence

of reactions for the synthesis of a given target compound and will employ protecting groups as necessary. This will depend inter alia on factors such as the nature of other functional groups present in a particular substrate. Clearly, the type of chemistry involved will influence the choice of reagent that is used in the said synthetic steps, the need, and type, of protecting groups that are employed, and the sequence for accomplishing the protection/deprotection steps. These and other reaction parameters will be evident to the skilled person by reference to standard textbooks and to the examples provided herein.

(59) Sensitive functional groups may need to be protected and deprotected during synthesis of a compound of the invention. This may be achieved by conventional methods, for example as described in "Protective Groups in Organic Synthesis" by TW Greene and PGM Wuts, John Wiley & Sons Inc. (1999), and references therein.

(60) Throughout this specification these abbreviations have the following meanings: TFAA—trifluoroacetic anhydride NaHMDS—Sodium bis(trimethylsilyl)amide DMSO—dimethylsulfoxide aq.—aqueous conc.—concentrated DCM—dichloromethane DMF—N,N-dimethylformamide h—hour quant.—quantitative HPLC—high performance liquid chromatography min—minute PE—petroleum ether r.t.—room temperature sat.—saturated TFA—trifluoroacetic acid THF—tetrahydrofuran

(61) Certain compounds of the invention can be accessed according to or analogously to the general synthetic schemes below. Certain compounds of the invention can be accessed according to or analogously to Examples 1 to 50 below.

(62) General Synthetic Schemes

(63) Compounds of formula I can be made according to schemes A to D. Alcohol A can be reacted with triphosgene (e.g. in the presence of pyridine and DCM at r.t.) to provide compounds of formula B (a subset of compounds of formula I).

(64) ##STR00028##

(65) Alcohol A can be reacted with thiophosgene (e.g. in the presence of pyridine and DCM at r.t.) to provide compounds of formula C (a subset of compounds of formula I).

(66) ##STR00029##

(67) Compounds of formula A can be made according to methods described in WO2019/141980 and according to Scheme C. Carboxylic acid D can be reacted with a ketone, e.g. acetone or benzophenone, (e.g. in the presence of TFA and TFAA) to provide ketal E. Reaction with an anion of amine F (e.g. the anion formed where F has been deprotonated with NaHMDS) under suitable conditions (e.g. THF at -40°C . to 0°C .) can provide compound A.

(68) ##STR00030##

is selected from Me and Ph.

Scheme C

(69) A number of routes to amine F can be envisaged. Certain of these are shown in Schemes D to H. Amino bromide G can be converted to thiourea H (e.g. using benzoylchloride, NH.sub.4SCN , Br.sub.2, in acetone at reflux followed by 10% aqueous NaOH solution at reflux) which can be converted to amine F e.g. using CuI, Cs.sub.2CO.sub.3, 1,10-phenanthroline in dioxane at reflux.

(70) ##STR00031##

(71) Thiourea J can be converted to amine F, e.g. using either Br.sub.2 in CHCl.sub.3 at 0°C . to reflux or Br.sub.2, AcOH and LiBr.

(72) ##STR00032##

(73) Thiocyanate K can be converted to amine F, e.g. using either H.sub.2, Pd/C in acetic acid at room temperature or Fe/AcOH at room temperature.

(74) ##STR00033##

(75) Bicycle L can be aminated to form amine F, e.g. using $\text{tBu.sub.2Zn(TMP)Li}$, THF at room temperature followed by copper cyanide and BnONH.sub.2 at room temperature.

(76) ##STR00034##

(77) Aminoethiol M can react with compound N to form amine F, e.g. in THF at reflux.

(78) ##STR00035##

(79) A subset of amine F, amines of formula R (in which x is an integer from 0 to 5) can be formed according to scheme I. Reaction of nitrofluoro compound O with phenol P in the presence of a base (e.g. K₂CO₃ or NaH in DMF) can provide ether Q. Reduction (e.g. with Pd/C and H₂ in ethanol at room temperature or Fe, NH₄Cl in THF/methanol at 60° C.) followed by reaction with KSCN or NaSCN (e.g. in the presence of Br₂ in AcOH or methanol at 0° C. to room temperature) can provide amines R (a subset of amines F).

(80) ##STR00036##

EXAMPLES

(81) General Methods

(82) Flash chromatography was carried out using a Biotage Isolera 4, with Biotage® SNAP KP-Sil cartridges, packed with 50 µm silica particles with a surface area of 500 m²/g, or alternative cartridges (e.g. Puriflash, produced by Interchim) where stated, or using silica gel (40-63 µm particles). Visualisation was carried out with UV light (254 nm) and by staining with either potassium permanganate, phosphomolybdic acid (PMA) or ninhydrin solutions.

(83) All ¹H NMR spectra were obtained on a Bruker AVIII 400 with 5 mm QNP or Bruker AVI 500 with 5 mm QNP. Chemical shifts are expressed in parts per million (δ) and are referenced to the solvent. Coupling constants J are expressed in Hertz (Hz).

(84) MS was carried out on a Waters Alliance ZQ MS, using a YMC-Triart C18 50×2 mm, 5 µm LC column (solvent: 5-90% gradient of acetonitrile in water (with 1% by volume of 28% (by weight) aqueous ammonia solution)) by Method A, or (solvent: 5-90% gradient of acetonitrile in water (with 1% formic acid)) by Method B. Flow rate: 0.8 mL/min. Wavelengths were 254 and 210 nm.

(85) Method a (5 Minute Basic pH)

(86) Column: YMC-Triart C18 50×2 mm, 5 µm. Flow rate: 0.8 mL/min. Injection volume: 5 µL.

(87) Mobile Phase A H₂O B CH₃CN C 50% H₂O/50% CH₃CN+1.0% ammonia (aq.)

(88) TABLE-US-00001 Time A B C (min) (%) (%) (%) 0 95 0 5 4.0 0 95 5 4.4 0 95 5 4.5 95 5 0 4.5 STOP

Method B (5 Minute Acidic pH)

(89) Column: YMC-Triart C18 50×2 mm, 5 µm. Flow rate: 0.8 mL/min. Injection volume: 5 µL.

(90) Mobile Phase A H₂O B CH₃CN C 50% H₂O/50% CH₃CN+1.0% formic acid

(91) TABLE-US-00002 Time A B C (min) (%) (%) (%) 0 95 0 5 4.0 0 95 5 4.4 0 95 5 4.5 95 5 0 4.5 STOP

(92) Alternatively, MS was carried on a Waters Acquity UPLC-QDA UV-MS system using Method C (high pH) or Method D (low pH):

(93) Method C (3.5 Minute Basic pH)

(94) Mobile phases: Water (A)/Acetonitrile (B) both with 0.1% (v/v) ammonia

(95) TABLE-US-00003 Flow rate Time % A % B (mL/min) Initial 98 2 1.0 0.2 98 2 1.0 2.5 2 98 1.0 3.0 2 98 1.0 3.1 98 2 1.0 3.5 98 2 1.0

(96) Column: BEH C18 2.1×50 mm, 1.7 µm @50° C.

(97) Method D (3.5 Minute Acidic pH)

(98) Mobile phases: Water (A)/Acetonitrile (B) both with 0.1% (v/v) formic acid

(99) TABLE-US-00004 Flow rate Time % A % B (mL/min) Initial 98 2 1.0 0.2 98 2 1.0 2.5 2 98 1.0 3.0 2 98 1.0 3.1 98 2 1.0 3.5 98 2 1.0

(100) Column: CSH C18 2.1×50 mm, 1.7 µm @50° C.

(101) All reagents were obtained from commercial suppliers and used as supplied unless otherwise stated.

(102) All compounds are named using ChemBioDraw Ultra 14.0.

Intermediate A: 2-chloro-1-(2-chlorophenoxy)-4-nitrobenzene

(103) ##STR00037##

(104) 3-Chloro-4-fluoronitrobenzene (4.0 g, 23 mmol) and 2-chlorophenol (2.36 mL, 22.8 mmol) were dissolved in DMF (10 mL). Potassium carbonate (3.15 g, 22.8 mmol) was added, and the reaction heated to 80° C. for 48 h. The suspension was allowed to cool to r.t. and water was added. The mixture was extracted with EtOAc and the combined extracts were dried (MgSO₄) and evaporated under vacuum to yield the title compound as a yellow solid (6.47 g, quant.). ¹H NMR δ (500 MHz, DMSO-d₆) 8.48 (d, J=2.8 Hz, 1H), 8.17 (dd, J=9.1, 2.8 Hz, 1H), 7.70 (dd, J=8.3, 1.6 Hz, 1H), 7.55-7.46 (m, 1H), 7.40 (ddd, J=8.3, 6.7, 1.6 Hz, 2H), 6.89 (d, J=9.1 Hz, 1H).

Intermediates B-I

(105) The following Intermediates were prepared using the general method described in Intermediate A from commercial materials.

(106) TABLE-US-00005 Intermediate No. Compound ¹H NMR/LCMS B 1-(2-methoxyethoxy)-2-(4-nitrophenoxy)benzene  ¹H NMR δ (500 MHz, DMSO-d₆) 8.21 (d, J = 9.2 Hz, 2H), 7.32-7.25 (m, 1H), 7.25-7.19 (m, 1H), 7.09-7.01 (m, 1H), 7.00 (d, J = 9.2 Hz, 2H), 6.79-6.67 (m, 1H), 4.10-4.03 (m, 2H), 3.47-3.36 (m, 2H), 3.32 (s, 3H); LCMS (Method B): 3.16 min (290.1, MH⁺). C 2-((2-chlorophenyl)thio)-5-nitropyridine  ¹H NMR δ (500 MHz, CDCl₃) 9.24 (dd, J = 2.7, 0.6 Hz, 1H), 8.26 (dd, J = 8.9, 2.7 Hz, 1H), 7.76 (dd, J = 7.7, 1.6 Hz, 1H), 7.64 (dd, J = 8.0, 1.4 Hz, 1H), 7.55-7.47 (m, 1H), 7.46-7.37 (m, 1H), 6.99 (dd, J = 8.9, 0.6 Hz, 1H); LCMS (Method A): 3.51 min (267.0, MH⁺). D 1-(4-nitrophenoxy)-2-(trifluoromethoxy)benzene  ¹H NMR δ (500 MHz, DMSO-d₆) 8.27 (d, J = 9.3 Hz, 2H), 7.62 (dt, J = 8.1, 1.3 Hz, 1H), 7.53 (td, J = 7.8, 1.6 Hz, 1H), 7.48-7.39 (m, 2H), 7.15 (d, J = 9.3 Hz, 2H). E 1-(cyclopropylmethoxy)-2-(4-nitrophenoxy)benzene  ¹H NMR δ (500 MHz, DMSO-d₆) 8.21 (d, J = 9.3 Hz, 2H), 7.30-7.24 (m, 1H), 7.21 (ddd, J = 8.2, 4.0, 1.6 Hz, 2H), 7.04 (dd, J = 6.9, 1.0 Hz, 1H), 7.00 (d, J = 9.3 Hz, 2H), 3.81 (m, 2H), 1.03-0.89 (m, 1H), 0.40-0.31 (m, 2H), 0.13-0.03 (m, 2H); LCMS (Method B): 3.64 min, no ionisation. F (2-methoxyphenyl)(4-nitrophenyl)sulfane  ¹H NMR δ (500 MHz, DMSO-d₆) 8.11 (d, J = 9.1 Hz, 2H), 7.61-7.43 (m, 2H), 7.25 (d, J = 1.0 Hz, 1H), 7.19 (d, J = 9.1 Hz, 2H), 7.14-7.04 (m, 1H), 3.78 (s, 3H); LCMS (Method B): 3.46 min, no ionisation. G 2-(2-chloro-4-(trifluoromethyl)phenoxy)-5-nitropyridine  ¹H NMR δ (500 MHz, DMSO-d₆) 9.04 (dd, J = 2.8, 0.5 Hz, 1H), 8.71 (dd, J = 9.1, 2.8 Hz, 1H), 8.13 (d, J = 1.9 Hz, 1H), 7.92-7.81 (m, 1H), 7.70 (d, J = 8.0 Hz, 1H), 7.50 (dd, J = 9.1, 0.5 Hz, 1H). H 1-ethoxy-2-(4-nitrophenoxy)benzene  ¹H NMR δ (500 MHz, DMSO-d₆) 8.21 (d, J = 9.2 Hz, 2H), 7.29 (t, J = 7.8 Hz, 1H), 7.21 (d, J = 8.2 Hz, 2H), 7.04 (d, J = 7.5 Hz, 1H), 7.00 (d, J = 9.2 Hz, 2H), 4.01 (q, J = 6.9 Hz, 2H), 1.10 (d, J = 6.9 Hz, 3H); LCMS (Method B): 4.10 min (260.1, MH⁺). I 1-(difluoromethoxy)-2-(4-nitrophenoxy)benzene  LCMS (Method B): 3.85 min (282.0, MH⁺).

Intermediate J: 2-chloro-N-methyl-N-(4-nitrophenyl)aniline

(107) ##STR00046##

(108) Sodium hydride (60% in mineral oil) (198 mg, 4.96 mmol) was suspended in DMF (5 mL) under nitrogen and cooled to 0° C. 2-Chloro-N-methylamine (0.436 mL, 3.54 mmol) was added, followed by 4-fluoronitrobenzene (0.50 g, 3.54 mmol) in DMF (5 mL). The reaction mixture was warmed to r.t. and stirred for 18 h. After addition of the reaction mixture to water (~25 mL), the suspension was stirred for 15 min and filtered. The solid was washed with water and dried under vacuum to yield the title compound as an orange solid (1.13 g, quant.). ¹H NMR δ (500 MHz, CDCl₃) 8.11 (d, J=9.5 Hz, 2H), 7.58 (dd, J=7.6, 1.9 Hz, 1H), 7.45-7.30 (m, 3H), 6.52 (d, J=9.1 Hz, 2H), 3.38 (s, 3H); LCMS (Method A): 3.65 min (263.1, MH⁺).

Intermediates K-P

(109) The following Intermediates were prepared using the general method described in Intermediate J from commercial materials.

(110) TABLE-US-00006 Intermediate No. Compound .sup.1H NMR/LCMS K 4-chloro-N-methyl-N-(4- LCMS (Method A): 3.75 min, (263.1, nitrophenyl)aniline MH.sup.+).  5-chloro-N-(4- LCMS (Method A): 2.95 min (250.0, nitrophenyl)pyridin-3-amine MH.sup.+).

 2-(2-chlorophenoxy)-5- nitropyridine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 9.03 (d, J = 2.5 Hz, 1H), 8.67 (dd, J = 9.1, 2.9 Hz, 1H), 7.64 (dd, J = 8.0, 1.5 Hz, 1H), 7.52-7.34 (m, 4H); LCMS (Method A): 3.12 min (250.9, MH.sup.+). N 2-chloro-3-((5-nitropyridin-2- yl)oxy)pyridine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 9.04 (dd, J = 2.8, 0.4 Hz, 1H), 8.71 (dd, J = 9.1, 2.9 Hz, 1H), 8.41 (dd, J = 4.7, 1.6 Hz, 1H), 7.99 (dd, J = 8.0, 1.6 Hz, 1H), 7.61 (dd, J = 8.0, 4.7 Hz, 1H), 7.49 (dd, J = 9.1, 0.4 Hz, 1H); LCMS (Method A): 2.79 min (251.9, MH.sup.+). O 2-(2-methoxyphenoxy)-5-nitropyridine  .sup.1H NMR δ .sub.H (500 MHz, CDCl.sub.3) 9.04 (d, J = 2.8 Hz, 1H), 8.48 (dd, J = 9.0, 2.8 Hz, 1H), 7.32 (dd, J = 8.2, 0.9 Hz, 1H), 7.22-7.16 (m, 1H), 7.11-7.01 (m, 3H), 3.78 (s, 3H); LCMS (Method B): 2.90 min (247.0, MH.sup.+). P 2-chloro-3-((4-nitrophenyl)thio)pyridine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.35 (dd, J = 4.7, 1.5 Hz, 1H), 8.31-8.23 (m, 2H), 8.02 (dd, J = 8.0, 1.5 Hz, 1H), 7.87-7.70 (m, 2H), 7.33 (dd, J = 8.0, 4.7 Hz, 1H); LCMS (Method A): 3.96 min (267.0, MH.sup.+).

Intermediate Q: 2-(cyclopropylmethoxy)phenol

(111) ##STR00053##

(112) A solution of 1,2-dihydroxybenzene (5.0 g, 45.4 mmol) in DMF (20 mL) was treated with potassium carbonate (7.53 g, 54.5 mmol) and (bromomethyl)cyclopropane (4.40 mL, 45.4 mmol) and stirred at 60° C. overnight. The reaction was allowed to cool to r.t. and then quenched with water and extracted with EtOAc. The organics were washed with brine, dried (MgSO.sub.4) and concentrated under reduced pressure to give an oil. The residue was purified by chromatography (SiO.sub.2, 0-5% EtOAc in PE) to yield the title compound as a pale yellow oil (2.2 g, 30%).

.sup.1H NMR δ .sub.H (500 MHz, CDCl.sub.3) δ 6.96-6.92 (m, 1H), 6.92-6.80 (m, 3H), 5.78 (s, 1H), 3.87 (d, J=7.1 Hz, 2H), 1.36-1.22 (m, 1H), 0.70-0.60 (m, 2H), 0.35 (q, J=5.2 Hz, 2H); LCMS (Method B): 2.54 min, no ionisation.

Intermediate R: 5-chloro-N-methyl-N-(4-nitrophenyl)pyridin-3-amine

(113) ##STR00054##

(114) An ice-cold solution of Intermediate L (754 mg, 3.02 mmol) in THF (10 mL) was treated with sodium hydride (60% in mineral oil) (145 mg, 3.62 mmol). The reaction was stirred for 15 min, then iodomethane (0.282 mL, 4.53 mmol) was added. The reaction mixture was stirred at r.t. for 2 h, then quenched with water and extracted with EtOAc. The organics were washed with brine, dried (MgSO.sub.4) and concentrated under reduced pressure to give the title compound as a red solid (773 mg, 97%). LCMS (Method A): 3.10 min (264.1, MH.sup.+).

Intermediate S: 3-chloro-4-(2-chlorophenoxy) aniline

(115) ##STR00055##

(116) Intermediate A (6.47 g, 23.5 mmol) was dissolved in EtOH (20 mL). A saturated aqueous solution of ammonium chloride (5 mL) was added to the reaction mixture along with iron (6.65 g, 117 mmol). The reaction mixture was heated at 80° C. for 18 h. After cooling to r.t., the mixture was filtered through Dicalite®, washing with EtOAc and the filtrate was concentrated under reduced pressure. The residue was partitioned between water and DCM. The organic extracts were dried (MgSO.sub.4) and concentrated under reduced pressure to give the title compound as a brown solid (5.37 g, 90%). .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 7.51 (dd, J=7.9, 1.5 Hz, 1H), 7.26-7.20 (m, 1H), 7.04 (td, J=7.9, 1.4 Hz, 1H), 6.91 (d, J=8.7 Hz, 1H), 6.74 (d, J=2.6 Hz, 1H), 6.61 (dd, J=8.3, 1.4 Hz, 1H), 6.57 (dd, J=8.7, 2.6 Hz, 1H), 5.38 (s, 2H); LCMS (Method A): 3.35 min (254.0, MH.sup.+).

Intermediates T-AH

(117) The following Intermediates were prepared using the general method described in Intermediate S from the appropriate intermediates.

(118) TABLE-US-00007 Intermediate No. Compound .sup.1H NMR/LCMS T 4-(2-(2-methoxyethoxy) phenoxy)aniline  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 7.08 (d, J = 1.5 Hz, 1H), 7.02-6.96 (m, 1H), 6.88-6.83 (m, 1H), 6.75-6.71 (m, 1H), 6.68 (d, J = 8.8 Hz, 2H), 6.55 (d, J = 8.8 Hz, 2H), 4.88 (s, 2H), 4.14- 4.06 (m, 2H), 3.65-3.57 (m, 2H), 3.27 (s, 3H); LCMS (Method B): 1.89 min (260.1, MH.sup.+). U 6-((2-chlorophenyl)thio)pyridin- 3-amine  .sup.1H NMR δ .sub.H (500 MHz, CDCl.sub.3) 8.14 (d, J = 2.9 Hz, 1H), 7.44-7.39 (m, 1H), 7.26-7.13 (m, 4H), 6.99 (dd, J = 8.4, 2.9 Hz, 1H), 4.00 (br s, 2H); LCMS (Method A): 2.73 min (237.0, MH.sup.+). V 4-(2-(trifluoromethoxy) phenoxy)aniline  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 7.49- 7.38 (m, 1H), 7.30 (ddd, J = 8.3, 7.6, 1.6 Hz, 1H), 7.18-7.05 (m, 1H), 6.84 (dd, J = 8.3, 1.6 Hz, 1H), 6.81-6.73 (m, 2H), 6.68-6.55 (m, 2H), 5.05 (s, 2H). W 4-(2-(cyclopropylmethoxy) phenoxy)aniline  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 7.03 (dd, J = 8.1, 1.6 Hz, 1H), 7.00-6.93 (m, 1H), 6.83 (td, J = 7.6, 1.6 Hz, 1H), 6.74 (dd, J = 8.0, 1.6 Hz, 1H), 6.67 (d, J = 8.8 Hz, 2H), 6.54 (d, J = 8.8 Hz, 2H), 4.87 (br s, 2H), 3.82 (d, J = 6.9 Hz, 1H), 3.16 (d, J = 5.2 Hz, 1H), 1.14 (m, 1H), 0.59-0.41 (m, 2H), 0.24 (m, 2H); LCMS (Method B): 2.33 min (256.1, MH.sup.+). X 4-((2-methoxyphenyl)thio) aniline  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) δ 7.13 (d, J = 8.5 Hz, 2H), 7.08-7.02 (m, 1H), 6.93 (dd, J = 8.1, 1.1 Hz, 1H), 6.77 (td, J = 7.6, 1.2 Hz, 1H), 6.62 (d, J = 8.6 Hz, 2H), 6.50 (dd, J = 7.8, 1.6 Hz, 1H), 3.81 (s, 3H), NH.sub.2 not observed; LCMS (Method B): 2.80 min (232.1, MH.sup.+). Y N1-(2-chlorophenyl)-N1- LCMS (Method A): 3.07 min (233.1 methylbenzene-1,4-diamine MH.sup.+).  Z N1-(4-chlorophenyl)-N1- LCMS (Method A): 3.38 min (233.1, methylbenzene-1,4-diamine MH.sup.+).  AA N1-(5-chloropyridin-3-yl)-N1- LCMS (Method A): 2.67 min (234.1, methylbenzene-1,4-diamine MH.sup.+).  AB 6-(2-chlorophenoxy)pyridin-3- amine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 7.52 (dd, J = 8.1, 1.6 Hz, 1H), 7.47 (dd, J = 2.9, 0.5 Hz, 1H), 7.32 (ddd, J = 8.4, 7.5, 1.6 Hz, 1H), 7.21-7.13 (m, 1H), 7.08 (ddd, J = 9.6, 8.4, 2.2 Hz, 2H), 6.82 (dd, J = 8.6, 0.5 Hz, 1H), 5.08 (s, 2H); LCMS (Method A): 2.54 min (221.0, MH.sup.+). AC 6-((2-chloropyridin-3-yl)oxy) pyridin-3-amine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.21 (dd, J = 4.6, 1.6 Hz, 1H), 7.59 (dd, J = 8.1, 1.6 Hz, 1H), 7.53-7.37 (m, 2H), 7.13 (dd, J = 8.6, 2.9 Hz, 1H), 6.90 (d, J = 8.6 Hz, 1H), 5.14 (s, 2H); LCMS (Method A): 1.95 min (222.0, MH.sup.+). AD 6-(2-methoxyphenoxy)pyridin- 3-amine  .sup.1H NMR δ .sub.H (500 MHz, CDCl.sub.3) 7.74 (d, J = 2.6 Hz, 1H), 7.20-7.13 (m, 2H), 7.10 (dd, J = 8.2, 1.5 Hz, 1H), 7.01 (dd, J = 8.2, 1.5 Hz, 1H), 7.00- 6.95 (m, 1H), 6.79 (d, J = 8.6 Hz, 1H), 3.81 (s, 3H), 3.81 (br s, 2H); LCMS (Method A): 2.12 min (217.0, MH.sup.+). AE 4-((2-chloropyridin-3-yl)thio)aniline  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.23 (dd, J = 4.7, 1.5 Hz, 1H), 7.82 (dd, J = 7.9, 1.5 Hz, 1H), 7.21-7.09 (m, 3H), 6.67-6.56 (m, 2H), 5.51 (s, 2H); LCMS (Method A): 3.01 min (237.0, MH.sup.+). AF 6-(2-chloro-4-(trifluoromethyl) phenoxy)pyridin-3-amine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) 7.97 (d, J = 1.8 Hz, 1H), 7.68 (ddd, J = 2.8, 1.8, 0.6 Hz, 1H), 7.52 (d, J = 2.6 Hz, 1H), 7.20 (d, J = 8.6 Hz, 1H), 7.14 (dd, J = 8.6, 2.9 Hz, 1H), 6.93 (d, J = 8.6 Hz, 1H), 5.24 (s, 2H); LCMS (Method A): 3.28 min (289.0, MH.sup.+). AG 4-(2-ethoxyphenoxy)aniline  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 7.05 (dd, J = 8.1, 1.6 Hz, 1H), 7.02- 6.96 (m, 1H), 6.83 (td, J = 7.6, 1.6 Hz, 1H), 6.74 (dd, J = 8.0, 1.6 Hz, 1H), 6.66 (d, J = 8.9 Hz, 2H), 6.54 (d, J = 8.9 Hz, 2H), 4.86 (s, 2H), 4.02 (q, J = 7.0 Hz, 2H), 1.26 (t, J = 7.0 Hz, 3H); LCMS (Method B): 2.56 min (231.0, MH.sup.+). AH 4-(2-(difluoromethoxy)phenoxy) aniline  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 7.26 (d, J = 8.0 Hz, 1H), 7.16 (td, J = 7.4, 6.7, 1.6 Hz, 2H), 7.05 (td, J = 7.8, 1.5 Hz, 1H), 6.81 (d, J = 1.5 Hz, 1H), 6.77 (dd, J = 8.7, 2.0 Hz, 2H), 6.59 (d,

J = 8.8 Hz, 2H), 4.99 (br s, 2H); LCMS (Method B): 2.90 min (252.0, MH.sup.+).

Intermediate AI: N6-(2-chlorophenyl)-N6-methylbenzo[d]thiazole-2,6-diamine

(119) ##STR00071##

(120) Sodium thiocyanate (575 mg, 7.09 mmol) was dissolved in MeOH (10 mL) and cooled to 0° C. Bromine (0.219 mL, 4.25 mmol) was added dropwise and the reaction stirred for 5 min.

Intermediate Y (825 mg, 3.55 mmol) in MeOH (10 mL) was added, and reaction mixture allowed to warm to r.t. and stirred for 48 h. The reaction was concentrated under reduced pressure and partitioned between DCM and sat. aq. NaHCO₃. The aq. layer was further extracted with DCM and the combined organics were dried (MgSO₄) and concentrated in vacuo. The residue was purified by chromatography (SiO₂, 0-50% EtOAc in PE) to yield the title compound as a grey foam (719 mg, 70%). ¹H NMR δ (500 MHz, DMSO-d₆) 7.55 (dd, J=8.0, 1.6 Hz, 1H), 7.43-7.38 (m, 17H), 7.34 (dd, J=8.0, 1.6 Hz, 1H), 7.30-7.24 (m, 1H), 7.14 (d, J=8.7 Hz, 3H), 6.98 (d, J=2.5 Hz, 1H), 6.45 (dd, J=8.7, 2.5 Hz, 1H), 3.19 (z, 3H); LCMS (Method A): 3.05 min (290.0 MH).

Intermediates AJ-AY

(121) The following Intermediates were prepared using the general method described in Intermediate A/from the appropriate intermediate or commercial material.

(122) TABLE-US-00008 Intermediate No. Compound ¹H NMR/LCMS AJ N6-(4-chlorophenyl)-N6-methylbenzo[d]thiazole-2,6-diamine  ¹H NMR δ (500 MHz, DMSO-d₆) 7.51 (d, J = 2.3 Hz, 1H), 7.45 (br s, 2H), 7.33 (d, J = 8.5 Hz, 1H), 7.21-7.16 (m, 2H), 7.01 (dd, J = 8.5, 2.3 Hz, 1H), 6.75-6.69 (m, 2H), 3.22 (s, 3H); LCMS (Method A): 3.25 min (290.0, MH.sup.+). AK N6-(5-chloropyridin-3-yl)-N6-methylbenzo[d]thiazole-2,6-diamine  ¹H NMR δ (500 MHz, DMSO-d₆) 7.94 (d, J = 2.5 Hz, 1H), 7.91 (d, J = 1.9 Hz, 1H), 7.62 (d, J = 2.2 Hz, 1H), 7.53 (br s, 2H), 7.38 (d, J = 8.5 Hz, 1H), 7.10 (dd, J = 8.5, 2.2 Hz, 1H), 7.08-7.04 (m, 1H), 3.27 (s, 3H); LCMS (Method A): 2.60 min (291.0, MH.sup.+). AL 7-chloro-6-(2-chlorophenoxy)benzo[d]thiazol-2-amine  Mixture of regioisomers LCMS (Method A): 3.26 min (311.0, 313.0 MH.sup.+ (CI isotopes)). 5-chloro-6-(2-chlorophenoxy)benzo[d]thiazol-2-amine  AM 6-(2-(2-methoxyethoxy)phenoxy)benzo[d]thiazol-2-amine  ¹H NMR δ (500 MHz, DMSO-d₆) 7.32 (s, 1H), 7.24 (d, J = 1.8 Hz, 1H), 7.15-7.06 (m, 2H), 6.91 (d, J = 3.6 Hz, 2H), 6.80 (dd, J = 8.6, 2.6 Hz, 1H), 4.15-4.03 (m, 2H), 3.60-3.47 (m, 2H), 3.19 (s, 3H). NH₂ not seen; LCMS (Method A): 2.25 min (317.0, MH.sup.+). AN 5-((2-chlorophenyl)thio)thiazolo[5,4-b]pyridin-2-amine  ¹H NMR δ (500 MHz, DMSO-d₆) 7.92 (br s, 2H), 7.62-7.57 (m, 2H), 7.40-7.29 (m, 3H), 7.15 (d, J = 8.3 Hz, 1H); LCMS (Method B): 2.68 min (294.0, MH.sup.+). AO 6-(2-(trifluoromethoxy)phenoxy)benzo[d]thiazol-2-amine  ¹H NMR δ (500 MHz, DMSO-d₆) 8.64 (s, 2H), 7.59-7.47 (m, 2H), 7.48-7.35 (m, 2H), 7.27 (td, J = 7.8, 1.5 Hz, 1H), 7.07 (td, J = 8.4, 2.0 Hz, 2H); LCMS (Method A): 3.35 min (327.0, MH.sup.+). AP 5-(2-chlorophenoxy)thiazolo[5,4-b]pyridin-2-amine  ¹H NMR δ (500 MHz, CDCl₃) 7.72 (dt, J = 8.6, 1.0 Hz, 1H), 7.51-7.41 (m, 1H), 7.35-7.24 (m, 2H), 7.17 (m, 1H), 6.88 (d, 1H), 5.76 (s, 2H); LCMS (Method A): 2.74 min (278.0, MH.sup.+). AQ 5-((2-chloropyridin-3-yl)oxy)thiazolo[5,4-b]pyridin-2-amine  ¹H NMR δ (500 MHz, DMSO-d₆) 8.31 (dd, J = 4.7, 1.6 Hz, 1H), 7.81 (dd, J = 8.0, 1.6 Hz, 1H), 7.77 (d, J = 8.6 Hz, 1H), 7.65 (s, 2H), 7.52 (dd, J = 8.0, 4.7 Hz, 1H), 7.05 (d, J = 8.6 Hz, 1H); LCMS (Method A): 2.21 min (279.0, MH.sup.+). AR 5-(2-methoxyphenoxy)thiazolo[5,4-b]pyridin-2-amine  ¹H NMR δ (500 MHz, DMSO-d₆) 7.67 (d, J = 8.6 Hz, 1H), 7.52 (br s, 2H), 7.21 (ddd, J = 8.2, 7.3, 1.6 Hz, 1H), 7.18-7.08 (m, 2H), 7.01-6.94 (m, 1H), 6.80 (d, J = 8.6 Hz, 1H), 3.70 (s, 3H); LCMS (Method A): 2.43 min (274.0, MH.sup.+). AS 6-(2-(cyclopropylmethoxy)phenoxy)benzo[d]thiazol-2-amine  ¹H NMR δ (500 MHz, DMSO-d₆) δ 7.32 (s, 1H), 7.22-7.26 (m,

2H), 7.11- 7.05 (m, 2H), 6.92-6.89 (m, 1H), 6.80 (dd, J = 8.7, 2.6 Hz, 1H), 3.82 (d, J = 6.8 Hz, 2H), 0.90-0.74 (m, 1H), 0.49-0.37 (m, 2H), 0.22-0.09 (m, 2H), NH.sub.2 not seen; LCMS (Method B): 2.76 min (313.1, MH.sup.+). AT 6-((2-methoxyphenyl)thio) benzo[d]thiazol-2-amine  1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) δ 8.57 (s, 2H), 7.84 (d, J = 1.6 Hz, 1H), 7.42 (d, J = 8.3 Hz, 1H), 7.33 (dd, J = 8.4, 1.9 Hz, 1H), 7.22 (ddd, J = 8.2, 7.3, 1.7 Hz, 1H), 7.05 (dd, J = 8.2, 1.0 Hz, 1H), 6.90-6.84 (m, 1H), 6.85- 6.73 (m, 1H), 3.83 (s, 3H). AU 6-isobutoxybenzo[d]thiazol-2- amine  1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) 7.27 (d, J = 2.6 Hz, 1H), 7.23-7.18 (m, 3H), 6.80 (dd, J = 8.7, 2.6 Hz, 1H), 3.71 (d, J = 6.6 Hz, 2H), 2.05-1.94 (m, 1H), 0.97 (d, J = 6.7 Hz, 6H); LCMS (Method B): 2.34 min (223.0, MH+). AV 6-((2-chloropyridin-3- yl)thio)benzo[d]thiazol-2-amine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) 8.23 (dd, J = 4.7, 1.5 Hz, 1H), 8.02-7.81 (m, 2H), 7.71 (s, 2H), 7.50-7.29 (m, 2H), 7.18 (dd, J = 7.9, 4.7 Hz, 1H); LCMS (Method A): 2.93 min (294.0, MH+). AW 5-(2-chloro-4-(trifluoromethyl) phenoxy)thiazolo[5,4-b]pyridin- 2-amine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) 8.05 (d, J = 1.7 Hz, 1H), 7.82-7.74 (m, 2H), 7.69 (s, 2H), 7.44 (d, J = 8.5 Hz, 1H), 7.08 (d, J = 8.5 Hz, 1H); LCMS (Method B): 3.62 min (346.0, MH+). AX 6-(2-ethoxyphenoxy) benzo[d]thiazol-2-amine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) 7.13 (dd, J = 4.9, 1.3 Hz, 1H), 7.05 (dd, J = 8.1, 1.6 Hz, 1H), 7.02-6.96 (m, 1H), 6.87-6.82 (m, 1H), 6.75 (dd, J = 8.1, 1.6 Hz, 1H), 6.68-6.65 (m, 1H), 6.57-6.54 (m, 1H), 4.87 (br s, 2H), 4.03 (q, J = 6.8 Hz, 2H), 1.27 (t, J = 7.0 Hz, 3H). AY 6-(2-(difluoromethoxy)phenoxy) benzo[d]thiazol-2-amine  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) 7.43 (br s, 1H), 7.39 (d, J = 2.5 Hz, 1H), 7.32 (d, J = 8.6 Hz, 2H), 7.22 (td, J = 7.8, 1.7 Hz, 1H), 7.15 (td, J = 7.7, 1.7 Hz, 1H), 6.95 (dd, J = 8.0, 1.6 Hz, 1H), 6.91 (dd, J = 8.7, 2.6 Hz, 1H), NH.sub.2 not seen; LCMS (Method B): 3.07 min (307.1, MH+).

Intermediate AZ: N-(6-((2-chlorophenyl)(methyl)amino)benzo[d]thiazol-2-yl)-3-hydroxy-4-methoxypicolinamide

(123) ##STR00089##

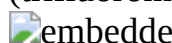
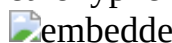
(124) A mixture of Intermediate AI (250 mg, 0.863 mmol) and 8-methoxy-2,2-dimethyl-4H-[1,3]dioxino[5,4-b]pyridin-4-one 2,2,2-trifluoroacetate (418 mg, 1.29 mmol) were dissolved in DMF (3 mL). NaHMDS solution (2M in THF) (2.16 mL, 4.31 mmol) was added, and the reaction was stirred at r.t. for 7 days. The reaction was quenched with water and stirred for 30 mins. The resulting solid was filtered, washed with water, slurried in EtOH and dried under vacuum to give the title compound as a brown solid (224 mg, 59%). .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) 7.63 (d, J=4.6 Hz, 1H), 7.59 (d, J=7.9 Hz, 1H), 7.49-7.36 (m, 3H), 7.35-7.28 (m, 1H), 7.15 (d, J=2.2 Hz, 1H), 6.74 (d, J=4.7 Hz, 1H), 6.54 (dd, J=8.8, 2.3 Hz, 1H), 3.76 (s, 3H), 3.25 (s, 3H). No NH/OH observed; LCMS (Method A): 1.88 min (441.0, MH.sup.+).

Intermediates BA-BP

(125) The following Intermediates were prepared using the general method described in Intermediate AZ from the appropriate intermediate.

(126) TABLE-US-00009 Intermediate No. Compound .sup.1H NMR/LCMS BA N-(6-((4-chlorophenyl)(methyl) amino)benzo[d]thiazol-2-yl)-3- hydroxy-4- methoxypicolinamide 0  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) 7.67 (d, J = 4.8 Hz, 1H), 7.64 (d, J = 2.1 Hz, 1H), 7.60 (d, J = 8.5 Hz, 1H), 7.22- 7.16 (m, 2H), 7.10 (dd, J = 8.6, 2.2 Hz, 1H), 6.81-6.74 (m, 3H), 5.74 (s, 1H), 3.76 (s, 3H), 3.26 (s, 3H). One NH/OH not observed; LCMS (Method A): 1.98 min (441.0, MH.sup.+). BB N-(6-((5-chloropyridin-3- yl) (methyl)amino)benzo[d] thiazol-2-yl)-3-hydroxy-4- methoxypicolinamide  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sub.6) 7.97 (d, J = 2.5 Hz, 1H), 7.91 (d, J = 1.9 Hz, 1H), 7.76 (d, J = 2.1 Hz, 1H), 7.70 (d, J = 4.8 Hz, 1H), 7.66 (d, J = 8.5 Hz, 1H), 7.19 (dd, J = 8.5, 2.2 Hz, 1H), 7.13-7.08 (m, 1H), 6.80 (d, J = 4.9 Hz, 1H), 5.75 (s, 1H), 3.77 (s, 3H). One OH/NH not observed. One CH.sub.3 peak obscured by water; LCMS (Method A): 1.65 min (442.0, MH.sup.+). BC N-(7-chloro-6-(2- chlorophenoxy)benzo[d]thiazol- 2-yl)-3-hydroxy-4-

methoxypicolinamide  embedded image. Mixture of regioisomers 1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 7.86- 7.74 (m, 1.5H), 7.70 (s, 0.5H), 7.59 (dd, J = 8.1, 5.6 Hz, 2H), 7.29 (t, J = 7.6 Hz, 2H), 7.18-7.08 (m, 2H), 6.88 (dd, J = 18.9, 5.0 Hz, 1H), 6.81 (t, J = 8.4 Hz, 1H), 3.81 (d, J = 6.7 Hz, 3H); LCMS (Method B): 3.00 min (462.0, MH.sup.+). N-(5-chloro-6-(2-chlorophenoxy)benzo[d]thiazol-2-yl)-3-hydroxy-4-methoxypicolinamide  embedded image BD 3-hydroxy-4-methoxy-N-(6-(2-(2-methoxyethoxy)phenoxy)benzo[d]thiazol-2-yl)picolinamide  embedded image .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 7.66 (d, J = 3.7 Hz, 1H), 7.56 (dd, J = 9.1, 1.3 Hz, 1H), 7.39-7.35 (m, 1H), 7.19- 7.10 (m, 2H), 6.99-6.91 (m, 3H), 6.76 (dd, J = 6.0, 1.3 Hz, 1H), 4.14- 4.09 (m, 2H), 3.76 (s, 3H), 3.57-3.51 (m, 2H), 3.19 (s, 3H), No NH/OH observed; LCMS (Method B): 2.43 min (468.1, MH.sup.+). BE N-(5-((2-chlorophenyl)thio)thiazolo[5,4-b]pyridin-2-yl)-3- hydroxy-4- methoxypicolinamide  embedded image .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 7.85 (t, J = 6.9 Hz, 2H), 7.65-7.55 (m, 2H), 7.44-7.29 (m, 4H), 7.15 (dd, J = 8.3, 4.6 Hz, 1H), 6.90 (d, J = 5.0 Hz, 1H), 3.81 (s, 3H); LCMS (Method A): 1.21 min (444.9, MH.sup.+). BF 3-hydroxy-4-methoxy-N-(6-(2- (trifluoromethoxy)phenoxy)benzo[d]thiazol-2-yl)picolinamide  embedded image .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 7.71 (d, J = 4.4 Hz, 1H), 7.64 (d, J = 8.7 Hz, 1H), 7.57 (d, J = 2.3 Hz, 1H), 7.51 (d, J = 8.4 Hz, 1H), 7.40-7.32 (m, 1H), 7.24-7.18 (m, 1H), 7.03 (dd, J = 8.6, 2.6 Hz, 2H), 6.80 (d, J = 4.7 Hz, 1H), 3.78 (s, 3H), No NH/OH observed; LCMS (Method A): 2.93 min (478.0, MH.sup.+). BG N-(5-(2- chlorophenoxy)thiazolo[5,4- b]pyridin-2-yl)-3-hydroxy-4- methoxypicolinamide  embedded image .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 7.94 (d, J = 8.6 Hz, 1H), 7.70 (d, J = 4.9 Hz, 1H), 7.55 (dd, J = 7.9, 1.5 Hz, 1H), 7.36-7.32 (m, 1H), 7.29-7.15 (m, 2H), 6.98 (d, J = 8.6 Hz, 1H), 6.78 (d, J = 5.0 Hz, 1H), 3.73 (s, 3H), No NH/OH observed; LCMS (Method A): 1.86 min (429.0, MH.sup.+). BH N-(5-((2-chloropyridin-3- yl)oxy)thiazolo[5,4-b]pyridin-2-yl)-3-hydroxy-4- methoxypicolinamide  embedded image .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 8.33 (dd, J = 4.7, 1.6 Hz, 1H), 8.04 (d, J = 8.6 Hz, 1H), 7.86 (dd, J = 8.0, 1.6 Hz, 1H), 7.79 (d, J = 4.9 Hz, 1H), 7.55 (dd, J = 8.0, 4.7 Hz, 1H), 7.14 (d, J = 8.6 Hz, 1H), 6.86 (d, J = 5.0 Hz, 1H), 3.80 (s, 3H), No NH/OH observed; LCMS (Method A): 1.58 min (430.0, MH.sup.+). BI 3-hydroxy-4-methoxy-N-(5-(2- methoxyphenoxy)thiazolo[5,4- b]pyridin-2-yl)picolinamide  embedded image .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 7.92 (d, J = 8.6 Hz, 1H), 7.73 (d, J = 4.9 Hz, 1H), 7.25-7.19 (m, 1H), 7.19-7.13 (m, 2H), 7.02-6.96 (m, 1H), 6.88 (d, J = 8.6 Hz, 1H), 6.82 (d, J = 5.0 Hz, 1H), 3.77 (s, 3H), 3.69 (s, 3H), No NH/OH observed; LCMS (Method A): 1.70 min (425.0, MH.sup.+). BJ 3-hydroxy-N-(6- isobutoxybenzo[d]thiazol-2-yl)- 4-methoxypicolinamide 00  embedded image 1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 7.63 (d, J = 4.8 Hz, 1H), 7.50 (d, J = 8.7 Hz, 1H), 7.42 (d, J = 2.5 Hz, 1H), 6.93 (dd, J = 8.8, 2.6 Hz, 1H), 6.73 (d, J = 4.8 Hz, 1H), 3.78 (d, J = 6.5 Hz, 2H), 3.75 (s, 3H), 2.09-1.97 (m, 1H), 1.00 (d, J = 6.7 Hz, 6H), No NH/OH observed; LCMS (Method B): 2.72 min (374.0, MH.sup.+). BK N-(6-(2-(cyclopropylmethoxy) phenoxy)benzo[d]thiazol-2-yl)- 3-hydroxy-4- methoxypicolinamide 01  embedded image .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 7.66 (d, J = 4.8 Hz, 1H), 7.56 (d, J = 8.7 Hz, 1H), 7.36 (d, J = 2.5 Hz, 1H), 7.14- 7.09 (m, 2H), 6.99 (d, J = 7.5 Hz, 1H), 6.96-6.90 (m, 2H), 6.75 (d, J = 4.9 Hz, 1H), 3.84 (d, J = 6.8 Hz, 2H), 3.75 (s, 3H), 1.14-1.03 (m, 1H), 0.46- 0.38 (m, 2H), 0.21-0.13 (m, 2H), No NH/OH observed; LCMS (Method B): 2.88 min (464.1, MH.sup.+). BL 3-hydroxy-4-methoxy-N-(6-((2- methoxyphenyl)thio)benzo[d] thiazol-2-yl)picolinamide 02  embedded image .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 7.87 (d, J = 1.7 Hz, 1H), 7.68 (d, J = 4.9 Hz, 1H), 7.58 (d, J = 8.3 Hz, 1H), 7.28 (dd, J = 8.3, 1.9 Hz, 1H), 7.16-7.04 (m, 1H), 6.97 (d, J = 7.2 Hz, 1H), 6.82- 6.72 (m, 2H), 6.66 (dd, J = 7.8, 1.6 Hz, 1H), 3.79 (s, 3H), 3.72 (s, 3H), No NH/OH observed; LCMS (Method A): 1.77 min (440.0, MH.sup.+). BM N-(6-((2-chloropyridin-3- yl)thio)benzo[d]thiazol-2-yl)-3- hydroxy-4- methoxypicolinamide 03  embedded image .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆.sup.6) 8.24 (ddd, J = 8.0, 4.7, 1.5 Hz, 1H), 8.04 (d, J = 1.6 Hz, 1H), 7.92-7.83 (m, 2H), 7.75 (d, J = 4.9 Hz, 1H), 7.71-7.65 (m, 1H), 7.44 (dd, J = 8.3, 1.8 Hz, 1H), 7.35 (dt, J = 8.3, 5.0 Hz, 1H), 7.21- 7.15 (m, 1H), 6.84 (d, J =

4.9 Hz, 1H), 3.80 (s, 3H); LCMS (Method B): 3.18 min (445.0, MH.sup.+). BA N-(5-(2-chloro-4-(trifluoromethyl)phenoxy)thiazolo [5,4-b]pyridin-2-yl)-3-hydroxy-4-methoxypicolinamide 04
 .sup.1H NMR δ.sub.H (500 MHz, DMSO-d.sup.6) 8.04- 7.93 (m, 2H), 7.76-7.68 (m, 2H), 7.43 (d, J = 8.5 Hz, 1H), 7.10 (d, J = 8.6 Hz, 1H), 6.80 (d, J = 5.0 Hz, 1H), 3.74 (s, 3H), No NH/OH observed; LCMS (Method A): 1.87 min (497.0, MH.sup.+). BO N-(6-(2-ethoxyphenoxy) benzo[d]thiazol-2-yl)-3- hydroxy-4- methoxypicolinamide 05
 .sup.1H NMR δ.sub.H (500 MHz, DMSO-d.sup.6) 7.68 (m, 1H), 7.61-7.52 (m, 1H), 7.40- 7.33 (m, 1H), 7.17-7.09 (m, 2H), 6.96 (m, 3H), 6.79 (m, 1H), 4.04 (m, 2H), 3.77 (s, 3H), 1.20 (m, 3H), No NH/OH observed; LCMS (Method B): 3.16 min (438.1, MH.sup.+). BP N-(6-(2-(difluoromethoxy) phenoxy)benzo[d]thiazol-2-yl)- 3-hydroxy-4- methoxypicolinamide 06
 .sup.1H NMR δ.sub.H (500 MHz, DMSO-d.sup.6) 7.69 (m, 1H), 7.65-7.59 (m, 1H), 7.53 (s, 1H), 7.32 (s, 1H), 7.27-7.13 (m, 2H), 7.00 (m, 2H), 6.93 (m, 1H), 6.78 (m, 1H), 3.77 (s, 3H), No NH/OH observed; LCMS (Method B): 3.07 min (460.1, MH.sup.+).

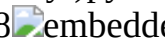
Example 1: 3-(6-(2-Chloro-4-fluorophenoxy)benzo[d]thiazol-2-yl)-8-methoxy-2H-pyrido[2,3-e][1,3]oxazine-2,4(3H)-dione

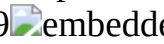
(127) ##STR00107##

(128) Triphosgene (53.2 mg, 0.179 mmol) was added to a solution of N-(6-(2-chloro-4-fluorophenoxy)benzo[d]thiazol-2-yl)-3-hydroxy-4-methoxypicolinamide (40 mg, 0.09 mmol) in DCM (1.5 mL). Pyridine (0.109 mL, 1.35 mmol) was added, and the reaction was stirred at r.t. for 30 min. The reaction mixture was purified by column chromatography (SiO.sub.2, 0-100% EtOAc in PE) and the title compound was isolated as a yellow gum (10 mg, 25%). .sup.1H NMR δ.sub.H(500 MHz, DMSO-d.sup.6) 8.58 (d, J=5.4 Hz, 1H), 8.10 (d, J=8.9 Hz, 1H), 7.75-7.69 (m, 2H), 7.59 (d, J=5.5 Hz, 1H), 7.38 (d, J=5.3 Hz, 1H), 7.33-7.29 (m, 2H), 4.06 (s, 3H); LCMS (Method B): 3.72 min (472.1, MH.sup.+).

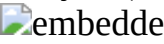
Examples 2-5

(129) The following Examples were prepared using the general method described in Example 1 from the appropriate Intermediate (preparation described above or in WO/2019/141980).

(130) TABLE-US-00010 Example No Compound .sup.1H NMR/LCMS 2 8-Methoxy-3-(6-((5-(trifluoromethyl)pyridin-3- yl)oxy)benzo[d]thiazol-2-yl)-2H- pyrido[2,3-e][1,3]oxazine-2,4(3H)-dione 08
 .sup.1H NMR δ.sub.H (500 MHz, DMSO-d.sup.6) 8.84 (br s, 1H), 8.80 (d, J = 2.6 Hz, 1H), 8.59 (d, J = 5.4 Hz, 1H), 8.17 (d, J = 8.9 Hz, 1H), 8.06-8.01 (m, 2H), 7.60 (d, J = 5.5 Hz, 1H), 7.48 (dd, J = 8.9, 2.6 Hz, 1H), 4.07 (s, 3H); LCMS (Method B): 3.36 min (489.1, MH.sup.+).

3 3-(6-(4- Fluorophenoxy)benzo[d]thiazol-2-yl)- 8-methoxy-2H-pyrido[2,3- e][1,3]oxazine-2,4(3H)-dione 09
 .sup.1H NMR δ.sub.H (500 MHz, DMSO-

d.sup.6) 8.58 (d, J = 5.4 Hz, 1H), 8.10 (d, J = 8.8 Hz, 1H), 7.81 (d, J = 2.5 Hz, 1H), 7.59 (d, J = 5.5 Hz, 1H), 7.34-7.27 (m, 3H), 7.22-7.18 (m, 2H), 4.06 (s, 3H); LCMS (Method B): 3.53 min (438.1, MH.sup.+).

4 8-Methoxy-3-(6- phenoxybenzo[d]thiazol-2-yl)-2H- pyrido[2,3-e][1,3]oxazine-2,4(3H)- dione 0
 .sup.1H NMR δ.sub.H (500 MHz, DMSO-d.sup.6) 8.58 (d, J = 5.4 Hz, 1H), 8.10 (d, J = 8.8 Hz, 1H), 7.85 (s, 1H), 7.58 (d, J = 5.4 Hz, 1H), 7.44 (t, J = 7.4 Hz, 2H), 7.32 (d, J = 8.8 Hz, 1H), 7.20 (t, J = 7.4 Hz, 1H), 7.12 (d, J = 8.2 Hz, 2H), 4.06 (s, 3H); LCMS (Method B): 3.48 min (420.1, MH.sup.+).

5 3-(6-(2- Chlorophenoxy)benzo[d]thiazol-2-yl)- 8-methoxy-2H-pyrido[2,3- e][1,3]oxazine-2,4(3H)-dione  .sup.1H NMR δ.sub.H (500 MHz, DMSO-d.sup.6) 8.66-8.61 (m, 1H), 8.00-7.96 (m, 1H), 7.55 (m, 2H), 7.23-7.14 (m, 2H), 7.09 (m, 3H), 3.94 (s, 3H); LCMS (Method B): 3.76 min (454, MH.sup.+).

Example 6: 3-(6-(2-Chloro-4-fluorophenoxy)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one

(131) ##STR00112##

(132) Thiophosgene (20.6 mg, 0.179 mmol) was added to a solution of N-(6-(2-chloro-4-fluorophenoxy)benzo[d]thiazol-2-yl)-3-hydroxy-4-methoxypicolinamide (40 mg, 0.09 mmol) in DCM (1 mL). Pyridine (0.109 mL, 1.35 mmol) was added, and the reaction was stirred at r.t. for 30

min. The reaction mixture was purified by column chromatography (SiO₂, 0-100% EtOAc in PE) and the title compound was isolated as a pale orange solid (22 mg, 50%). ¹H NMR δ (500 MHz, DMSO-d₆) 8.61 (d, J=5.4 Hz, 1H), 8.09 (d, J=8.8 Hz, 1H), 7.74-7.69 (m, 2H), 7.62 (d, J=5.5 Hz, 1H), 7.40 (dd, J=9.1, 5.3 Hz, 1H), 7.37-7.31 (m, 1H), 7.30 (dd, J=8.9, 2.6 Hz, 1H), 4.09 (s, 3H); LCMS (Method B): 4.05 min (488.1, MH⁺).

Examples 7-50

(133) The following Examples were prepared using the general method described in Example 6 from the appropriate Intermediate (preparation described above or in WO/2019/141980).

(134) TABLE-US-00011 Example No. Compound ¹H NMR/LCMS 7 3-(6-((5-Chloropyridin-3-yl)oxy)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  ¹H NMR δ (500 MHz, DMSO-d₆) 8.61 (d, J = 5.4 Hz, 1H), 8.50 (dd, J = 5.9, 2.1 Hz, 2H), 8.14 (d, J = 8.8 Hz, 1H), 7.99 (d, J = 2.4 Hz, 1H), 7.86-7.83 (m, 1H), 7.62 (d, J = 5.5 Hz, 1H), 7.43 (dd, J = 8.8, 2.3 Hz, 1H), 4.09 (s, 3H); LCMS (Method D): 2.04 min (471.0, MH⁺). 8 8-Methoxy-2-thioxo-3-(6-((5-(trifluoromethyl)pyridin-3-yl)oxy)benzo[d]thiazol-2-yl)-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one

 ¹H NMR δ (500 MHz, DMSO-d₆) 8.84 (br s, 1H), 8.82 (d, J = 2.6 Hz, 1H), 8.61 (d, J = 5.4 Hz, 1H), 8.16 (d, J = 8.8 Hz, 1H), 8.08-8.03 (m, 1H), 8.02 (d, J = 2.5 Hz, 1H), 7.63 (d, J = 5.5 Hz, 1H), 7.46 (dd, J = 8.9, 2.6 Hz, 1H), 4.09 (s, 3H); LCMS (Method D): 2.08 min (505.0, MH⁺). 9 3-(6-(3-Chlorophenoxy)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one 

¹H NMR δ (500 MHz, DMSO-d₆) 8.61 (d, J = 5.4 Hz, 1H), 8.12 (d, J = 8.9 Hz, 1H), 7.93 (d, J = 2.5 Hz, 1H), 7.62 (d, J = 5.5 Hz, 1H), 7.48-7.44 (m, 1H), 7.36 (dd, J = 8.9, 2.5 Hz, 1H), 7.27 (d, J = 8.0 Hz, 1H), 7.25-7.22 (m, 1H), 7.10 (dd, J = 8.3, 1.7 Hz, 1H), 4.09 (s, 3H); LCMS (Method B): 3.45 min (470.0, MH⁺). 10 3-(6-(4-Fluorophenoxy)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one 

¹H NMR δ (500 MHz, DMSO-d₆) 8.61 (d, J = 5.3 Hz, 1H), 8.09 (d, J = 8.8 Hz, 1H), 7.81 (s, 1H), 7.62 (d, J = 5.4 Hz, 1H), 7.37-7.25 (m, 3H), 7.25-7.14 (m, 2H), 4.09 (s, 3H); LCMS (Method B): 3.23 min (454.0, MH⁺). 11 3-(6-(3-Fluorophenoxy)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one 

¹H NMR δ (500 MHz, DMSO-d₆) 8.61 (d, J = 5.4 Hz, 1H), 8.12 (d, J = 8.9 Hz, 1H), 7.94 (d, J = 2.5 Hz, 1H), 7.62 (d, J = 5.5 Hz, 1H), 7.50-7.44 (m, 1H), 7.36 (dd, J = 8.9, 2.6 Hz, 1H), 7.07-7.03 (m, 2H), 6.99-6.93 (m, 1H), 4.09 (s, 3H); LCMS (Method B): 3.25 min (454.0, MH⁺). 12 3-(6-(2-Chloro-4-fluorophenoxy)-5-(2-chlorophenyl)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one 

¹H NMR δ (500 MHz, DMSO-d₆) 8.61 (d, J = 5.3 Hz, 1H), 8.04 (s, 1H), 7.69 (s, 1H), 7.63-7.54 (m, 3H), 7.47-7.42 (m, 2H), 7.34-7.30 (m, 1H), 7.30-7.24 (m, 2H), 4.09 (s, 3H); LCMS (Method B): 3.86 min (598.0, MH⁺). 13 3-(6-(2-Chloro-4-fluorophenoxy)-4-phenylbenzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one 

¹H NMR δ (500 MHz, DMSO-d₆) 8.59 (d, J = 5.3 Hz, 1H), 7.80-7.75 (m, 2H), 7.75-7.69 (m, 2H), 7.62-7.58 (m, 1H), 7.52-7.27 (m, 6H), 4.07 (s, 3H); LCMS (Method B): 3.96 min (564.0, MH⁺). 14 3-(5-(2-Chloro-4-fluorophenoxy)thiazolo[5,4-d]pyrimidin-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one 

¹H NMR δ (500 MHz, DMSO-d₆) 9.47 (s, 1H), 8.62 (d, J = 5.4 Hz, 1H), 7.71 (dd, J = 8.4, 3.0 Hz, 1H), 7.67-7.58 (m, 2H), 7.42-7.36 (m, 1H), 4.09 (s, 3H); LCMS (Method B): 3.11 min (489.9, MH⁺). 15 Isopropyl 2-(8-methoxy-4-oxo-2-thioxo-2H-pyrido[2,3-e][1,3]oxazin-3(4H)-yl)benzo[d]thiazole-6-carboxylate 

pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.60 (d, J = 5.3 Hz, 1H), 7.81 (d, J = 9.0 Hz, 1H), 7.68 (dd, J = 8.6, 2.8 Hz, 1H), 7.61 (d, J = 5.4 Hz, 1H), 7.54 (dd, J = 8.7, 5.8 Hz, 1H), 7.41-7.35 (m, 1H), 7.35-7.29 (m, 1H), 6.71 (dd, J = 9.0, 2.2 Hz, 1H), 4.08 (s, 3H), 3.29 (s, 3H); LCMS (Method B): 3.40 min (501.0, MH.sup.+). 17 8-Methoxy-3-(6- phenoxybenzo[d]thiazol-2-yl)-2- thioxo-2,3-dihydro-4H-pyrido[2,3- e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.60 (d, J = 5.4 Hz, 1H), 8.08 (d, J = 9.1 Hz, 1H), 7.83 (d, J = 2.5 Hz, 1H), 7.61 (d, J = 5.5 Hz, 1H), 7.44 (dd, J = 8.6, 7.4 Hz, 2H), 7.30 (dd, J = 8.9, 2.6 Hz, 1H), 7.20 (t, J = 7.4 Hz, 1H), 7.13 (dd, J = 8.3, 1.4 Hz, 2H), 4.07 (s, 3H); LCMS (Method D): 2.14 min (436.1, MH.sup.+). 18 3-(6-(2-Chlorophenoxy)benzo[d]thiazol-2-yl)- 8-methoxy-2-thioxo-2,3-dihydro-4H- pyrido[2,3-e][1,3]oxazin-4-one  1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.60 (dd, J = 5.4, 2.2 Hz, 1H), 8.09 (dd, J = 8.9, 1.8 Hz, 1H), 7.74 (t, J = 2.2 Hz, 1H), 7.66 (d, J = 8.2 Hz, 1H), 7.61 (dd, J = 5.4, 1.9 Hz, 1H), 7.43 (t, J = 7.0 Hz, 1H), 7.29 (q, J = 8.2 Hz, 3H), 4.08 (s, 3H); LCMS (Method B) 3.32 min (469.9/472.0, MH.sup.+). 19 3-(6-Benzoylbenzo[d]thiazol-2-yl)-8- methoxy-2-thioxo-2,3-dihydro-4H- pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.75 (s, 1H), 8.67 (d, J = 1.1 Hz, 1H), 8.61 (d, J = 5.4 Hz, 1H), 8.24 (d, J = 8.5 Hz, 1H), 7.96 (dd, J = 8.5, 1.8 Hz, 1H), 7.84 (dd, J = 8.2, 1.1 Hz, 2H), 7.75-7.70 (m, 2H), 7.61 (d, J = 6.9 Hz, 1H), 4.08 (s, 3H); LCMS (Method B): 3.01 min (448.0, MH.sup.+). 20 8-Methoxy-2-thioxo-3-(6-(2- (trifluoromethyl)phenoxy)benzo[d] thiazol-2-yl)-2,3-dihydro-4H- pyrido[2,3- e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.60 (d, J = 5.4 Hz, 1H), 8.12 (d, J = 8.9 Hz, 1H), 7.91 (d, J = 2.6 Hz, 1H), 7.84 (d, J = 7.9 Hz, 1H), 7.70 (d, J = 8.5 Hz, 1H), 7.61 (d, J = 5.5 Hz, 1H), 7.39 (d, J = 7.9 Hz, 1H), 7.36-7.32 (m, 1H), 7.23 (d, J = 8.5 Hz, 1H), 4.08 (s, 3H); LCMS (Method D): 2.17 min (503.9, MH.sup.+). 21 3-(6-(3,5- Difluorophenoxy)benzo[d]thiazol-2- yl)-8-methoxy-2-thioxo-2,3-dihydro- 4H-pyrido[2,3- e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.61 (d, J = 5.4 Hz, 1H), 8.14 (d, J = 8.9 Hz, 1H), 8.00 (d, J = 2.5 Hz, 1H), 7.63 (d, J = 5.5 Hz, 1H), 7.40 (dd, J = 8.9, 2.6 Hz, 1H), 7.33 (s, 1H), 7.08 (tt, J = 9.4, 2.3 Hz, 1H), 6.92 (dd, J = 8.5, 2.2 Hz, 1H), 4.09 (s, 3H); LCMS (Method D): 2.16 min (472.0, MH.sup.+). 22 3-(6-(Cyclohexyloxy)benzo[d]thiazol- 2-yl)-8-methoxy-2-thioxo-2,3-dihydro- 4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.59 (d, J = 5.5 Hz, 1H), 7.93 (d, J = 8.9 Hz, 1H), 7.79 (d, J = 2.6 Hz, 1H), 7.60 (d, J = 5.5 Hz, 1H), 7.16 (dd, J = 8.9, 2.6 Hz, 1H), 4.42 (s, 1H), 4.07 (s, 3H), 1.74 (m, 2H), 1.59-1.22 (m, 8H); LCMS (Method D): 2.24 min (442.1, MH.sup.+). 23 3-(6- (Cyclopentyloxy)benzo[d]thiazol-2- yl)-8-methoxy-2-thioxo-2,3-dihydro- 4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.61 (s, 1H), 7.95 (s, 2H), 7.59 (s, 2H), 4.94 (m, 1H), 4.08 (s, 3H), 1.84-1.69 (m, 2H), 1.63 (m, 4H), 0.88-0.74 (m, 2H); LCMS (Method D): 2.14 min (436.1, MH.sup.+). 24 3-(6-Benzylbenzo[d]thiazol-2-yl)-8- methoxy-2-thioxo-2,3-dihydro-4H- pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.59 (d, J = 5.5 Hz, 1H), 8.06 (s, 1H), 7.99 (d, J = 8.4 Hz, 1H), 7.60 (d, J = 5.5 Hz, 1H), 7.47 (dd, J = 8.4, 1.7 Hz, 1H), 7.32-7.28 (m, 4H), 7.21 (m, 1H), 4.13 (s, 2H), 4.07 (s, 3H); LCMS (Method B): 3.23 min (434.0, MH.sup.+). 25 3-(6-(4- Chlorophenoxy)benzo[d]thiazol-2-yl)- 8-methoxy-2-thioxo-2,3-dihydro-4H- pyrido[2,3-e][1,3]oxazin-4-one  LCMS (Method B): 3.46 min 469.9/472.0. 26 3-(6-(2- Chlorophenoxy)benzo[d]thiazol-2-yl)- 8-methoxy-2H-pyrido[2,3- e][1,3]oxazine-2,4(3H)-dithione  1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.57 (d, J = 5.4 Hz, 1H), 8.05 (d, J = 8.9 Hz, 1H), 7.71 (d, J = 2.6 Hz, 1H), 7.66 (dd, J = 8.0, 1.5 Hz, 1H), 7.59 (d, J = 5.4 Hz, 1H), 7.43 (td, J = 8.0, 1.5 Hz, 1H), 7.32-7.24 (m, 3H), 4.07 (s, 3H); LCMS (Method B): 3.46 min (485.9/487.9, MH.sup.+). 27 8-Methoxy-3-(6-(phenylthio)benzo[d]thiazol-2-yl)-2- thioxo-2,3-dihydro-4H-pyrido[2,3- e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.60 (d, J = 5.4 Hz, 1H), 8.23 (d, J = 1.6 Hz, 1H), 8.07 (d, J = 8.5 Hz, 1H), 7.61 (d, J = 5.5 Hz, 1H), 7.52 (dd, J = 8.5, 1.9

Hz, 1H), 7.42 (d, J = 4.5 Hz, 4H), 7.36 (dd, J = 9.4, 4.1 Hz, 1H), 4.08 (s, 3H); LCMS (Method B): 3.44 min (453.1, MH.sup.+). 28 3-(6-(2-Chloro-4-fluorobenzyl)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.61 (d, J = 5.4 Hz, 1H), 8.02 (d, J = 8.4 Hz, 1H), 7.99 (s, 1H), 7.62 (d, J = 5.5 Hz, 1H), 7.52-7.44 (m, 3H), 7.28-7.21 (m, 1H), 4.26 (s, 2H), 4.08 (s, 3H); LCMS (Method B): 3.56 min (485.9, MH.sup.+). 29 3-(6-(2-Chlorobenzyl)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.61 (d, J = 5.3 Hz, 1H), 8.04-7.99 (m, 2H), 7.62 (d, J = 5.4 Hz, 1H), 7.51-7.41 (m, 3H), 7.38-7.28 (m, 2H), 4.27 (s, 2H), 4.08 (s, 3H); LCMS (Method B): 3.44 min (468.0, MH.sup.+). 30 8-methoxy-3-(6-(2-methoxyphenoxy)benzo[d]thiazol-2-yl)-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.59 (d, J = 5.4 Hz, 1H), 8.00 (d, J = 8.9 Hz, 1H), 7.60 (d, J = 5.5 Hz, 1H), 7.57 (d, J = 2.6 Hz, 1H), 7.29-7.20 (m, 2H), 7.20-7.12 (m, 2H), 7.06-6.98 (m, 1H), 4.07 (s, 3H), 3.76 (s, 3H); LCMS (Method B): 3.09 min (466.0, MH.sup.+). 31 3-(6-((2-chlorophenyl)(methyl)amino)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.60 (d, J = 5.4 Hz, 1H), 7.81 (d, J = 9.0 Hz, 1H), 7.69-7.64 (m, 1H), 7.61 (d, J = 5.5 Hz, 1H), 7.52-7.46 (m, 2H), 7.44-7.37 (m, 1H), 7.35 (d, J = 2.5 Hz, 1H), 6.69 (dd, J = 9.1, 2.5 Hz, 1H), 4.08 (s, 3H), 3.32 (s, 3H); LCMS (Method B): 3.21 min (483.1, MH.sup.+). 32 3-(6-((4-chlorophenyl)(methyl)amino)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.61 (d, J = 5.4 Hz, 1H), 7.95 (d, J = 8.9 Hz, 1H), 7.82 (d, J = 2.4 Hz, 1H), 7.62 (d, J = 5.5 Hz, 1H), 7.40-7.33 (m, 2H), 7.24 (dd, J = 8.9, 2.4 Hz, 1H), 7.16-7.10 (m, 2H), 4.09 (s, 3H), 3.37 (s, 3H); LCMS (Method B): 3.38 min (483.0, MH.sup.+). 33 3-(6-((5-chloropyridin-3-yl)(methyl)amino)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.62 (d, J = 5.4 Hz, 1H), 8.24 (d, J = 2.5 Hz, 1H), 8.12 (d, J = 2.0 Hz, 1H), 8.09-8.03 (m, 2H), 7.62 (d, J = 5.5 Hz, 1H), 7.52-7.48 (m, 1H), 7.43 (dd, J = 8.8, 2.4 Hz, 1H), 4.09 (s, 3H), 3.42 (s, 3H); LCMS (Method B): 2.71 min (484.1, MH.sup.+). 34 3-(6-((2-chlorophenyl)thio)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.62 (d, J = 5.4 Hz, 1H), 8.35 (d, J = 1.8 Hz, 1H), 8.15 (d, J = 8.5 Hz, 1H), 7.61 (m, 3H), 7.38-7.31 (m, 2H), 7.20-7.13 (m, 1H), 4.09 (s, 3H); LCMS (Method B): 3.45 min (486.0, MH.sup.+). 35 3-(7-chloro-6-(2-chlorophenoxy)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  Unseparable mixture of regioisomers. .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.62 (dd, J = 8.1, 5.4 Hz, 1H), 8.41 (s, 0.6H), 8.12 (d, J = 8.9 Hz, 0.4H), 7.85 (s, 0.6H), 7.70-7.66 (m, 1H), 7.63 (dd, J = 9.2, 5.5 Hz, 1H), 7.42 (ddd, J = 12.8, 9.4, 1.6 Hz, 1H), 7.34-7.28 (m, 1H), 7.25 (d, J = 8.9 Hz, 0.4H), 7.22 (dd, J = 8.1, 1.5 Hz, 0.6H), 7.17 (dd, J = 8.2, 1.5 Hz, 0.4H), 4.09 (d, J = 5.4 Hz, 3H); LCMS (Method B): 3.67 min and 3.76 min (504.0, MH.sup.+). 36 3-(5-chloro-6-(2-chlorophenoxy)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  36 3-(6-((2-chloropyridin-3-yl)oxy)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.61 (dd, J = 5.4, 2.3 Hz, 1H), 8.32 (d, J = 2.3 Hz, 1H), 8.13 (d, J = 8.9 Hz, 1H), 7.87 (d, J = 2.5 Hz, 1H), 7.75 (d, J = 8.1 Hz, 1H), 7.62 (d, J = 5.3 Hz, 1H), 7.52 (dd, J = 8.1, 4.6 Hz, 1H), 7.43-7.34 (m, 1H), 4.09 (s, 3H); LCMS (Method B): 2.84 min (471.0, MH.sup.+). 37 8-methoxy-3-(6-(2-(2-methoxyethoxy)phenoxy)benzo[d]thiazol-2-yl)-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one  .sup.1H NMR δ .sub.H (500 MHz, DMSO-d₆) 8.61 (d, J = 5.4 Hz, 1H), 8.02 (d, J = 9.0 Hz, 1H), 7.63-7.59 (m, 2H), 7.26-7.22 (m, 2H), 7.22-7.16 (m, 2H), 7.06-7.02 (m, 1H), 4.13-4.10 (m, 2H), 4.09 (s, 3H), 3.50-3.47 (m, 2H), 3.11 (s, 3H). 38 3-(5-((2-chlorophenyl)thio)thiazolo[5,4-b]pyridin-2-

yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.61 (d, J = 5.4 Hz, 1H), 8.41 (d, J = 8.7 Hz, 1H), 7.83 (dd, J = 7.7, 1.6 Hz, 1H), 7.75 (dd, J = 8.0, 1.3 Hz, 1H), 7.60 (ddd, J = 9.7, 8.1, 3.6 Hz, 2H), 7.51 (td, J = 7.6, 1.4 Hz, 1H), 7.31 (d, J = 8.7 Hz, 1H), 4.09 (s, 3H); LCMS (Method B): 3.99 min (486.9, MH.sup.+).

39 8-methoxy-2-thioxo-3-(6-(2-(trifluoromethoxy)phenoxy)benzo[d]thiazol-2-yl)-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.60 (d, J = 5.4 Hz, 1H), 8.10 (d, J = 8.9 Hz, 1H), 7.85 (d, J = 2.5 Hz, 1H), 7.61 (d, J = 5.5 Hz, 1H), 7.60-7.56 (m, 1H), 7.46 (ddd, J = 8.1, 7.6, 1.6 Hz, 1H), 7.38-7.27 (m, 3H), 4.08 (s, 3H); LCMS (Method B): 3.99 min (520.0, MH.sup.+).

40 3-(5-(2-chlorophenoxy)thiazolo[5,4-b]pyridin-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.62 (d, J = 1.1 Hz, 1H), 8.60 (d, J = 2.3 Hz, 1H), 7.66 (d, J = 7.8 Hz, 1H), 7.62 (d, J = 5.5 Hz, 1H), 7.48 (d, J = 3.9 Hz, 2H), 7.44 (d, J = 8.9 Hz, 1H), 7.39-7.34 (m, 1H), 4.08 (s, 3H); LCMS (Method B): 3.23 min (471.0, MH.sup.+).

41 3-(5-((2-chloropyridin-3-yl)oxy)thiazolo[5,4-b]pyridin-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.67 (d, J = 5.3 Hz, 1H), 8.41 (d, J = 8.8 Hz, 1H), 8.38 (dd, J = 4.5, 1.3 Hz, 1H), 7.69 (dd, J = 8.0, 1.6 Hz, 1H), 7.40 (dd, J = 8.0, 4.7 Hz, 1H), 7.32 (d, J = 8.8 Hz, 1H), 7.23 (d, J = 5.4 Hz, 1H), 4.12 (s, 3H); LCMS (Method B): 2.79 min (472.0, MH.sup.+).

42 8-methoxy-3-(5-(2-methoxyphenoxy)thiazolo[5,4-b]pyridin-2-yl)-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.60 (m, 1H), 8.52 (m, 1H), 7.61 (m, 1H), 7.33-7.27 (m, 2H), 7.26-7.18 (m, 2H), 7.09-6.99 (m, 1H), 4.08 (s, 3H), 3.73 (s, 3H); LCMS (Method B): 3.00 min (467.0, MH.sup.+).

43 3-(6-isobutoxybenzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.60 (d, J = 5.4 Hz, 1H), 7.95 (d, J = 9.0 Hz, 1H), 7.77 (d, J = 2.6 Hz, 1H), 7.61 (d, J = 5.5 Hz, 1H), 7.19 (dd, J = 9.0, 2.6 Hz, 1H), 4.08 (s, 3H), 3.86 (d, J = 6.6 Hz, 2H), 2.15-2.03 (m, 1H), 1.02 (d, J = 6.7 Hz, 6H); LCMS (Method B): 3.29 min (416.0, MH.sup.+).

44 3-(6-(2-(cyclopropylmethoxy)phenoxy)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.60 (d, J = 5.4 Hz, 1H), 8.01 (d, J = 8.9 Hz, 1H), 7.64-7.57 (m, 2H), 7.25-7.14 (m, 4H), 7.06-6.97 (m, 1H), 4.08 (s, 3H), 3.85 (d, J = 6.8 Hz, 2H), 1.05-1.00 (m, 1H), 0.44-0.34 (m, 2H), 0.19-0.07 (m, 2H); LCMS (Method B): 4.05 min (506.1, MH.sup.+).

45 8-methoxy-3-(6-((2-methoxyphenyl)thio)benzo[d]thiazol-2-yl)-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.61 (d, J = 5.4 Hz, 1H), 8.13 (dd, J = 1.9, 0.5 Hz, 1H), 8.05 (dd, J = 8.6, 0.4 Hz, 1H), 7.62 (d, J = 5.6 Hz, 1H), 7.45 (dd, J = 8.6, 1.9 Hz, 1H), 7.38 (ddd, J = 8.3, 7.4, 1.7 Hz, 1H), 7.20-7.11 (m, 2H), 6.98 (td, J = 7.5, 1.2 Hz, 1H), 4.09 (s, 3H), 3.84 (s, 3H); LCMS (Method B): 3.90 min (482.0, MH.sup.+).

46 8-methoxy-2-thioxo-3-(6-(4-(trifluoromethyl)phenoxy)benzo[d]thiazol-2-yl)-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.61 (d, J = 5.4 Hz, 1H), 8.15 (d, J = 8.8 Hz, 1H), 8.01 (d, J = 2.5 Hz, 1H), 7.78 (d, J = 8.5 Hz, 2H), 7.62 (d, J = 5.5 Hz, 1H), 7.41 (dd, J = 8.9, 2.5 Hz, 1H), 7.28 (d, J = 8.4 Hz, 2H), 4.09 (s, 3H); LCMS (Method B): 4.17 min (504.0, MH.sup.+).

47 3-(6-((2-chloropyridin-3-yl)thio)benzo[d]thiazol-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.62 (d, J = 5.4 Hz, 1H), 8.49 (d, J = 1.3 Hz, 1H), 8.30 (dd, J = 4.7, 1.5 Hz, 1H), 8.17 (d, J = 8.9 Hz, 1H), 7.96 (dd, J = 8.0, 1.5 Hz, 1H), 7.73 (dd, J = 8.5, 1.8 Hz, 1H), 7.63 (d, J = 5.5 Hz, 1H), 7.25 (dd, J = 8.0, 4.7 Hz, 1H), 4.10 (s, 3H); LCMS (Method B): 3.95 min (486.9, MH.sup.+).

48 3-(5-(2-chloro-4-(trifluoromethyl)phenoxy)thiazolo[5,4-b]pyridin-2-yl)-8-methoxy-2-thioxo-2,3-dihydro-4H-pyrido[2,3-e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.66 (d, J = 8.9 Hz, 1H), 8.61 (d, J = 5.4 Hz, 1H), 8.14 (s, 1H), 7.88 (d, J = 8.4 Hz, 1H),

7.74 (d, J = 8.5 Hz, 1H), 7.62 (d, J = 5.5 Hz, 1H), 7.54 (d, J = 8.9 Hz, 1H), 4.08 (s, 3H); LCMS (Method B): 4.23 min (539.0, MH.sup.+). 49 3-(6-(2-ethoxyphenoxy) benzo[d]thiazol-2-yl)-8-methoxy-2- thioxo-2,3-dihydro-4H-pyrido[2,3- e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.61 (m, 1H), 8.02 (m, 1H), 7.64- 7.59 (m, 2H), 7.26- 7.15 (m, 4H), 7.02 (m, 1H), 4.09 (s, 3H), 4.08- 4.03 (m, 2H), 1.18 (t, J = 7.0 Hz, 3H); LCMS (Method B): 3.87 min (480.0, MH.sup.+). 50 3-(6-(2-(difluoromethoxy)phenoxy) benzo[d]thiazol-2-yl)-8-methoxy-2- thioxo-2,3-dihydro-4H-pyrido[2,3- e][1,3]oxazin-4-one .sup.1H NMR δ .sub.H (500 MHz, DMSO-d.sup.6) 8.61 (d, J = 5.4 Hz, 1H), 8.09 (d, J = 9.0 Hz, 1H), 7.77 (d, J = 2.5 Hz, 1H), 7.62 (d, J = 5.5 Hz, 1H), 7.45- 7.40 (m, 1H), 7.36-7.24 (m, 4H), 7.23 (s, 1H), 4.09 (s, 3H); LCMS (Method B): 3.74 min (502.1, MH.sup.+).

Example 51: Testing the Fungicidal Activity of the Compounds of the Invention

(135) Compounds were screened in 96 well plates with 10 compounds per plate. Each compound was screened using agar amended to 20, 2, 0.2 and 0.02 ppm of the test material. Proline at 50 and 10 ppm and 0.2% DMSO were used respectively as positive and negative controls. Each test concentration and standard were tested twice on a plate.

(136) Compounds were screened against the following three fungal pathogens—*Botrytis cinerea*, *Alternaria alternata* and *Zymoseptoria tritici*. The agar used in the test varied depending on the pathogen with Medium N used for *B. cinerea* and *A. alternata* and 1% potato dextrose agar for *Z. tritici*. For each pathogen sufficient spores were added to the appropriate agar to give 1,000 spores/mL agar of *A. alternata*, 5,000 spores/ml agar of *B. cinerea* and 10,000 spores/mL agar of *Z. tritici*.

(137) A $\times 10$ stock solution in 2% DMSO was produced for each dose i.e. 200, 20, 2 and 0.2 ppm, and 10 μ l of this added to the appropriate wells on the plate. An equivalent amount of 2% DMSO and Proline stock at 500 and 100 ppm were added for the controls. To each well, 90 μ l of the appropriate agar spore suspension was added to give the final well concentrations outlined in the first paragraph.

(138) Plates were incubated at room temperature (18° C.) and assessed after: a) 3 to 4 days *A. alternata* and *B. cinerea* b) 7 days for *Z. tritici*

(139) The amount of fungal growth in each well was compared to the DMSO controls and scored according to the following key: A—EC₅₀<2 ppm B—2 $\leq$ EC₅₀<20 C—EC₅₀ $\geq$ 20 D—No activity detected at the highest dose tested NT—Not tested

(140) The ranking in the table is:

(141) TABLE-US-00012 Example *Botrytis Alternaria Zymoseptoria* 1 B B A 2 B B A 3 B B A 4 B B A 5 B B A 6 B B A 7 B B A 8 B B A 9 B B A 10 B A A 11 B B B 12 NT NT A 13 NT NT A 14 NT NT A 15 NT NT A 16 NT NT A 17 B B A 18 NT NT A 19 NT NT B 20 B B A 21 B B A 22 B C B 23 B C A 24 NT NT B 25 NT NT A 26 NT NT A 27 C B A 28 B B A 29 C C A 30 NT NT A 31 NT NT A 32 NT NT A 33 NT NT B 34 NT NT A 35 NT NT A 36 C C A 37 NT NT B 38 C C A 39 C C A 40 C C A 41 C C C 42 C C B 43 NT NT A 44 NT NT B 45 NT NT B 46 NT NT A 47 C C A 48 C C A 49 C C A 50 C C A

Claims

1. A compound of formula I, or an agronomically acceptable salt or N-oxide thereof:

##STR00158## X^{sup.1} and X^{sup.2} are each independently selected from carbon and nitrogen; Y^{sup.1} is independently selected from O and S; =Y^{sup.2} is independently selected from =O and =S; Z^{sup.1} is independently absent or is selected from C(O)O, OC(O), O, S, S(O), S(O).sub.2, C(O)NR^{sup.5}, NR^{sup.5}C(O), S(O).sub.2NR^{sup.5}, NR^{sup.5}S(O).sub.2, S(O)NR^{sup.5}, NR^{sup.5}S(O), CR^{sup.6}R^{sup.7}, C(O), C(S), C=NOR^{sup.8}, C.sub.1-C.sub.3-alkylene and NR^{sup.5}; R^{sup.1} and R^{sup.12} are each independently at each occurrence selected from C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.3-C.sub.6-cycloalkyl, halogen, nitro, OR^{sup.9},

SR.sup.10, OS(O).sub.2R.sup.10, S(O).sub.2R.sup.10, C(O)OR.sup.10, C(O)NR.sup.10R.sup.10, C(O)R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, and NR.sup.10R.sup.11; R.sup.2 is independently at each occurrence selected from C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.3-C.sub.6-cycloalkyl, halogen, nitro, OR.sup.9, SR.sup.10, OS(O).sub.2R.sup.10, S(O).sub.2R.sup.10, C(O)OR.sup.10, C(O)NR.sup.10R.sup.10, C(O)R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, NR.sup.10R.sup.11 and phenyl optionally substituted with from 1 to 5 R.sup.12 groups; R.sup.3 is independently at each occurrence selected from C.sub.3-C.sub.8-alkyl and C.sub.0-C.sub.3-alkylene-R.sup.3a; wherein R.sup.3a is independently at each occurrence selected from phenyl, 5- or 6-membered heteroaryl, 5-, 6-, 7- or 8-membered heterocycloalkyl and C.sub.3-C.sub.8-cycloalkyl; said heterocycloalkyl or cycloalkyl group being monocyclic or bicyclic; said heteroaryl or phenyl group being optionally substituted with from 1 to 5 R.sup.12 groups or said heterocycloalkyl or cycloalkyl group being optionally substituted with from 1 to 4 R.sup.13 groups; said heterocycloalkyl or cycloalkyl group being optionally fused to phenyl or a 5- or 6-membered heteroaryl, said heteroaryl or phenyl group being optionally substituted with from 1 to 4 R.sup.12 groups; R.sup.4 is independently at each occurrence selected from: C.sub.1-C.sub.6-alkyl, C.sub.3-C.sub.6-cycloalkyl and C.sub.1-C.sub.6-haloalkyl; R.sup.5 and R.sup.10 are each independently at each occurrence selected from: H, C.sub.3-C.sub.6-cycloalkyl, C.sub.1-C.sub.6-alkyl and benzyl; or where two R.sup.10 groups are attached to the same nitrogen atom, said R.sup.10 groups, together with said nitrogen atom form a 4-, 5-, 6- or 7-membered heterocycloalkyl ring; R.sup.6 is independently at each occurrence selected from: H, C.sub.3-C.sub.6-cycloalkyl, C.sub.1-C.sub.6-alkyl, phenyl and 5- or 6-membered heteroaryl; R.sup.7 is independently at each occurrence selected from: H, halo and OR.sup.8; R.sup.8 is each independently at each occurrence selected from: H, C.sub.3-C.sub.6-cycloalkyl, C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.3-alkylene-R.sup.8a; wherein R.sup.8a is independently at each occurrence selected from phenyl and 5- or 6-membered heteroaryl; R.sup.9 is independently at each occurrence selected from: H, C.sub.1-C.sub.6-alkyl, C.sub.0-C.sub.3-alkylene-C.sub.3-C.sub.6-cycloalkyl, C(O)—C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.6-haloalkyl; R.sup.11 is independently at each occurrence selected from: H, C.sub.1-C.sub.6-alkyl, C(O)—C.sub.1-C.sub.6-alkyl and S(O).sub.2—C.sub.1-C.sub.6-alkyl; or where an R.sup.10 group and an R.sup.11 group are attached to the same nitrogen atom, said R.sup.10 and R.sup.11 groups, together with said nitrogen atom form a 4-, 5-, 6- or 7-membered heterocycloalkyl ring; R.sup.13 is independently at each occurrence selected from: =O, =S, C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.3-C.sub.6-cycloalkyl; halogen, nitro, OR.sup.9, SR.sup.10, OS(O).sub.2R.sup.10, S(O).sub.2R.sup.10, S(O).sub.2NR.sup.10R.sup.10, S(O)(NR.sup.10)R.sup.10, S(O)R.sup.10, cyano, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, and NR.sup.10R.sup.11; y is independently an integer selected from 0, 1 and 2; m is independently an integer selected from 0, 1, 2 and 3; and wherein any aforementioned alkyl, alkylene, alkenyl, cycloalkyl, heterocycloalkyl (including where two R.sup.10 groups or an R.sup.10 group and an R.sup.11 group together with a nitrogen to which they are attached form a heterocycloalkyl ring), alkynyl, C(O)-alkyl, S(O).sub.2-alkyl and benzyl is optionally substituted, where chemically possible, by 1 to 4 substituents which are each independently selected at each occurrence from the group consisting of: =O; =NR.sup.a, =NOR.sup.a, C.sub.1-C.sub.4-alkyl, halo, nitro, cyano, C.sub.1-C.sub.4-haloalkyl, C.sub.2-C.sub.4-alkenyl, C.sub.2-C.sub.4-alkynyl, NR.sup.aR.sup.b, S(O).sub.2R.sup.a, S(O)R.sup.a, S(O)(NR.sup.a)R.sup.a, S(O).sub.2NR.sup.aR.sup.a, CO.sub.2R.sup.a, C(O)R.sup.a, CONR.sup.aR.sup.a, OR.sup.a and SR.sup.a; wherein R.sup.a is independently selected from H and C.sub.1-C.sub.4-alkyl; and R.sup.b is independently H, C.sub.1-C.sub.4-alkyl, C(O)—C.sub.1-C.sub.4-alkyl, S(O).sub.2—C.sub.1-C.sub.4-alkyl.

2. A compound of claim 1, wherein R.sup.4 is independently selected from: C.sub.1-C.sub.6-alkyl

and C.sub.3-C.sub.6-cycloalkyl.

3. A compound of claim 1, wherein y is 0.

4. A compound of claim 1, wherein Y.sup.1 is O.

5. A compound of claim 1, wherein =Y.sup.2 is =S.

6. A compound of claim 1, wherein X.sup.1 is carbon.

7. A compound of claim 6, wherein X.sup.2 is carbon.

8. A compound of claim 6, wherein X.sup.2 is nitrogen.

9. A compound of claim 1, wherein Z.sup.1 is independently at each occurrence selected from O, S, C.sub.1-alkylene and NR.sup.5, wherein R.sup.5 is C.sub.1-C.sub.3-alkyl.

10. A compound of claim 9, wherein Z.sup.1 is O.

11. A compound of claim 1, wherein R.sup.3 is R.sup.3a.

12. A compound of claim 1, wherein R.sup.3a is optionally substituted phenyl.

13. A compound of claim 1, wherein R.sup.3 is C.sub.3-C.sub.8-alkyl.

14. A compound of claim 1, wherein m is 0.

15. A compound of claim 1, wherein the compound of formula (I) is selected from: ##STR00159##
##STR00160## ##STR00161## ##STR00162## ##STR00163## ##STR00164## ##STR00165##

16. A method for controlling fungal diseases, the method comprising applying an agronomically effective and substantially non-phytotoxic quantity of a compound of claim 1 to seeds of plants, to plants themselves or to an area where it is intended that plants will grow.

17. A fungicidal composition comprising an effective and non-phytotoxic amount of an active compound of claim 1.
